

A novel bi-level optimization scheduling framework for electricity-carbon-water Nexus in power systems[☆]

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HIGHLIGHTS

- An electricity-carbon-water nexus for real-time operation of the power system is established.
- A bi-level optimization strategy for source-load-storage (SLS) coordination is proposed.
- The water-carbon responsibilities of the power system are allocated.
- Low-carbon and water-saving tasks in the power system can be carried out simultaneously.

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ABSTRACT

The sustainable development of water-efficient and low-carbon power systems has become an increasingly urgent priority. Existing research primarily utilizes provincial-level power trade data to model the nexus between virtual water and carbon emissions. However, these studies often lack an integrated approach to power flow analysis, limiting their effectiveness in guiding power scheduling decisions. Furthermore, optimization efforts in power systems have focused primarily on carbon reduction, overlooking the need for a holistic optimization that addresses both water conservation and carbon reduction within the electricity-water-carbon nexus. To address these challenges, this paper proposes a bi-level scheduling model for power systems that synchronously optimizes water conservation, carbon reduction, and economic efficiency. The model integrates the coordination of generation, load, and energy storage. In the upper-level optimization, the model determines the optimal generation strategy for water and carbon savings. By accounting for carbon emission flow and virtual water flow, the power grid operator calculates electricity-water-carbon coupling prices at load nodes, based on water and carbon intensities. In the lower-level optimization, load aggregators (LAs) and energy storage operators respond to price signals by formulating optimal load-shifting and energy storage charging/discharging scheduling. To reduce computational complexity, the lower-level problem is transformed into Karush-Kuhn-Tucker (KKT) optimality conditions using the Lagrange multiplier method, enabling the bi-level optimization to be converted into a single-level problem. This is then solved using the Branch-and-Bound method to determine the optimal power flow for the system. Case studies using the improved IEEE 33-node model demonstrate that the proposed optimization model reasonably allocates the water consumption and carbon emission responsibilities to each node, relieving water stress in water-scarce areas, reducing the system's total water consumption and total carbon emissions, reducing wind and photovoltaic curtailment compared to the existing methods. Particularly, the model reduces the system's total water consumption by 29.64% and carbon emissions by 10.68% compared to Scenario 1. It supports the sustainable development of power systems by leveraging flexibility resources on both the load and storage.

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1. Introduction

The large-scale construction of transmission lines in China's power system has significantly enhanced the nationwide interconnectivity and large-scale scheduling capabilities of the power grid. However, it has also caused a spatial shift in the environmental pressures associated with power generation. Among these, water scarcity and excessive carbon emissions have emerged as major environmental challenges. The power sector, as the core of the energy system, consumes significant amounts of water and generates substantial carbon emissions during electricity production. From 2010 to 2016, the water consumption for electricity generation in China increased from 2.7 km³ to 6.5 km³, while carbon dioxide emissions rose from 507 million tons to 642 million tons [1]. This has had adverse effects on global climate change and regional water resources, underscoring the importance and urgency of addressing water consumption and carbon emissions in power systems [2].

In future power systems, conflicts may arise between carbon reduction and water resource conservation. For instance, its power mix of the Japan's nationally determined contributions (NDCs) could reduce existing carbon emissions by 50%, but water consumption would increase by 36% [3]. Similarly, in 2016, China's power transmission resulted in a net carbon emission reduction of 155 million tons but also caused a net increase in water consumption of 3.4 km³ [1]. Addressing this dual challenge of reducing carbon emissions while managing water resources effectively has become a critical issue in power system planning.

The concept of virtual water offers a novel perspective for analyzing water resource consumption in power systems. First introduced by Allan, virtual water initially referred to the water required to produce agricultural commodities [4,5]. Over time, it has evolved to represent the water resources embedded in the production of any goods or services [6]. In the context of power systems, virtual water refers to the water consumed in electricity generation [7]. In power systems, water consumed during electricity generation is transferred through the grid as virtual water, forming a virtual water transfer nexus [8]. Integrating the virtual water trade strategy with electricity-water relationships can mitigate regional water resource imbalances and promote a more equitable distribution of water resources [9,10]. Yan et al. use an interregional input-output model to account for virtual water transfer in electricity systems, exploring the relationship between regional electricity prices and water resources. Their findings show that in water-scarce but energy-rich provinces, electricity generation exacerbates water stress [11]. They emphasize the need to evaluate whether virtual water transfer via power transmission alleviates or exacerbates national water shortages, proposing a rationality evaluation model from the energy-water nexus perspective [12]. Objective indicators such as the Water Stress Index (WSI) and the concept of virtual water scarcity have significantly influenced the analysis of interregional virtual water transfer [13,14]. Zhang et al. construct an interprovincial virtual water transfer nexus and propose the concept of mixed virtual water content to quantify the water consumption coefficient per unit of electricity generation for each province [14]. They further develop an impact assessment system to evaluate the "positive" or "negative" impacts of electricity trade, offering insights for optimizing the national electricity trade structure. Although these studies have contributed significantly to understanding the interregional virtual water transfer and its impacts on water stress, they primarily focus on macro-level virtual water analysis, lacking integration with power system scheduling and control. As a result, they fall short in providing actionable insights for optimizing water savings within power system operations. Therefore, there is an urgent need to develop more refined methods for calculating virtual water flow that are directly applicable to power system management.

Similarly, the analysis of implicit carbon is equally critical in the low-carbon transition of power systems. Like virtual water, implicit carbon refers to the carbon emissions directly or indirectly generated during the production of a product [15]. In power systems, the total carbon

emissions resulting from electricity production are defined as implicit carbon. Most studies on implicit carbon focus on allocating carbon emission responsibilities, analyzing carbon emissions at national, regional, or provincial scales [16]. These studies typically derive conclusions from the consumption of primary energy sources, which limits their applicability at the power system level. To response, Kang et al. combine network analysis techniques to develop a theoretical framework and computational methods for carbon emission flows in power systems, though they did not account for grid losses [17,18]. Building on this, they propose a recursive algorithm for carbon emission flows based on node carbon intensity and adjacency characteristics [19]. Other researchers have adopted the carbon emission flow theory as a basis for optimizing low-carbon scheduling in power systems, distributing carbon emission responsibilities between generation and load sides [20–22]. Yu et al. develop carbon emission flow models for loads and energy storage devices and proposed a low-carbon scheduling strategy based on the carbon flow model at the storage side [23]. Their results demonstrate that the optimal scheduling approach reduces wind curtailment, decreases the output of high-emission units, and effectively lowers overall system carbon emissions while maintaining economic efficiency. Moreover, these studies integrate carbon emissions with power flow, providing effective means for carbon emission accounting and optimization in power systems. However, they often ignore the water resource constraints at the power flow level. As a result, they fail to simultaneously address carbon reduction and the varying water resource conditions across different regions.

Currently, power industry planning primarily emphasizes low-carbon development, but the integration of water conservation objectives with carbon reduction targets remains unclear. In response, literature develops an optimization model aimed at mitigating water resource scarcity risks in China's power sector and reducing carbon emissions [24]. It calculates the regional generation mix, associates virtual water flows, and carbon emissions under three different scenarios prioritizing water savings and carbon reductions, offering policy recommendations. This research provides a synergistic perspective by improving regional water-carbon performance based on changes in provincial-level generation structures. However, the study is limited by its reliance on historical provincial-level generation data, which fails to reflect specific conditions of individual power plants. Consequently, its results can only inform overall provincial-level generation management without offering insights into unit-level generation strategies for operational power plants. Zhang et al. incorporate carbon neutrality targets and water withdrawal constraints, constructing an enhanced SWITCH-CHINA capacity expansion and scheduling model with plant-level data resolution [25]. Their study explores the long-term transformation of China's power industry from 2020 to 2050, finding that achieving carbon neutrality by 2060 largely depends on the widespread deployment of carbon capture and storage (CCS) technologies in coal-fired plants in China's water-scarce northwest region after 2035. However, these studies primarily adopt a macro perspective, disconnected from power system load flow analysis [26–29]. As a result, they fail to deeply integrate virtual water and implicit carbon with power system scheduling. The lack of modeling and computational methods for integrating virtual water and carbon emissions flow into power system load flow analysis leaves the research on electricity-water-carbon coupling relationships incomplete.

Above literature review demonstrates that electricity, water resources, and carbon emissions are intricately linked, forming a complex relationship of synergy or trade-offs. However, the published power system optimizations often neglect the comprehensive consideration of water-saving and carbon-reduction objectives. This oversight could constrain the low-carbon transition of existing power systems due to water resource scarcity or exacerbate water-carbon conflicts, intensity jeopardizing the achievement of dual carbon targets. Motivated by these challenges, this paper presents a bi-level scheduling model for power systems that simultaneously optimizes water conservation, carbon

reduction, and economic efficiency. The key contributions of this study are as follows:

(1) This paper extends the research focus from the macro level to the operational power system level. Based on the life-cycle water consumption coefficients and carbon emission coefficients of China's major generation types, the study calculates the virtual water and implicit carbon of electricity. Utilizing power system load flow analysis, it tracks and analyzes the flows of implicit carbon and virtual water in the electricity grid from temporal and spatial dimensions.

(2) A electricity-carbon-water nexus model is established to account for the internal flow of virtual water and implicit carbon within the power system. Water consumption and carbon emission responsibilities are allocated to the load side through node water intensity and node carbon intensity calculations. This approach refines the study to align with the actual grid structure of power systems, providing a more granular understanding of the transmission of virtual water and implicit carbon within the power grid.

(3) The design considers a low-carbon, water-saving economic scheduling strategy that coordinates the source-load-storage (SLS) system. On the source side, it mitigates local water pressure and reduces carbon emissions by adjusting the output of power plants. On the load-storage side, a price-based demand response model is constructed using node carbon potential and node water potential in the power system's electricity-carbon-water nexus.

(4) A bi-level multi-objective optimization scheduling model for electricity-carbon-water nexus in power systems is further established based on the source-load-storage coordinated low-carbon and water-saving economic scheduling strategy. The upper layer mainly focuses on the economic scheduling of the power generation side, while the lower layer addresses the low-carbon and water-saving scheduling of the load-storage side. The Karush-Kuhn-Tucker (KKT) method is applied to transform the bi-level multi-objective model, and the branch-and-bound method is used to solve the optimal power flow of the power system. This approach refines the low-carbon and water-saving economic benefits of the power system. By increasing the response of flexible loads, the dependency on source-side adjustments is reduced, thus achieving coordinated low-carbon and water-saving development in the power system.

2. Power system electricity-carbon-water coupling nexus model

2.1. System structure analysis

The study begins by extending the carbon emission flow theory proposed by Kang et al. [17]. The concept behind carbon emission flow theory treats carbon emissions as a virtual network flow attached to power flow. It integrates carbon emissions with power flow analysis. In other words, the implicit carbon flow in a power system is a virtual quantity that moves along with the electricity transmission in the system, representing the transfer of carbon emission responsibilities. This implicit carbon flow starts at the power plants, flows through the grid following the power system's energy flow, and eventually reaches the load nodes, as shown in Fig. 1.

On the surface, carbon dioxide (CO_2) is emitted by power plants. However, in reality, CO_2 is consumed by the load side of the power system through implicit carbon flow. CO_2 is a greenhouse gas produced during energy generation by power plants, and along with CO_2 , other greenhouse gases such as sulfur dioxide (SO_2) and nitrogen oxides (NO_x) are also emitted. The process of generating electricity also consumes raw materials such as coal, oil, and water. Regardless of whether they are gases emitted or raw materials consumed, these are tangible, quantifiable substances. Therefore, it is reasonable to extend the carbon flow theory to account for the consumption or emission of other substances by power plants.

The material flow in a power system focuses on analyzing the production, transfer, and consumption of a specific substance within the power system, which is associated with the power flow in the system. Therefore, the calculation method for power system material flow is based on the results of power system load flow analysis. Material flow calculations are carried out using a proportional-sharing tracking method, based on the steady-state equilibrium distribution of the system's power flow. The material flow tracking method is suitable for power systems of all sizes. In medium and large power systems, a recursive algorithm can be employed instead of the direct algorithm, which requires generating and operating on large-scale matrices. This approach significantly reduces the computational complexity associated with large power systems [19]. In the following, the carbon emission flow theory in power systems is extended to material flow theory, with corresponding material flow indicators and calculation methods proposed for power systems. On this basis, the corresponding virtual water flow indicators and calculation methods are introduced.

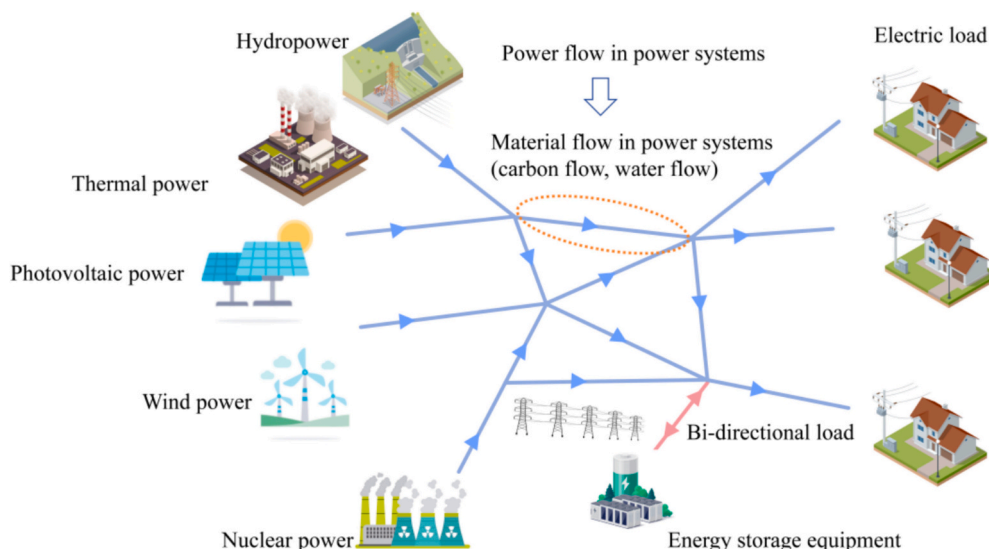


Fig. 1. Power system flow and virtual material flow model.

2.2. Implicit material flow theory

2.2.1. Branch implicit material flow

First, the concept of implicit material flow, denoted as F , is introduced to represent the magnitude of a particular substance flow along a branch. The branch implicit material flow is defined as the accumulated amount of implicit material flowing through a specific branch during a given time period, which accompanies the transmission of electricity. Its dimension is the same as that of the analyzed material's consumption or emission.

When only one end of a branch is connected to a power plant, the implicit material flow value in that branch is equal to the corresponding substance's consumption or emission amount. Conversely, when one end of the branch is only connected to a load, all implicit material flows in that branch will flow toward the load. Like power flow, branch implicit material flow has a directional nature.

In this paper, F_c is used to represent the branch implicit carbon flow, having the same dimension as carbon emissions (ton). F_w represents the branch virtual water flow, with the same dimension as water consumption (m^3).

2.2.2. Branch implicit material flow rate

The branch implicit material flow rate is defined as the implicit material flow passing through a branch per unit of time, denoted as v . In this paper, v_c represents the branch implicit carbon flow rate (ton/s), and v_w represents the branch implicit material flow rate (m^3/s). The branch implicit material flow rate is numerically equal to the time derivative of the branch implicit material flow F , and they are related as follows:

$$v(t) = \frac{dF(t)}{dt} \quad (1)$$

where $v(t)$ represents the implicit material flow through the branch.

2.2.3. Branch implicit material flow density

To better illustrate the coupling relationship between power flow and the consumption or emission of materials in the power system, the ratio of the branch implicit material flow rate to the active power flow of the branch as the branch implicit material flow density, denoted by ρ , as shown in eq. (2). Particularly, ρ_c represents the branch implicit carbon flow density (ton/kWh), and ρ_w represents the branch virtual water flow density (m^3/kWh).

$$\rho = \frac{v}{P} \quad (2)$$

For lines leaving the power plant, the branch implicit material flow density is equal to the consumption or emission intensity of the implicit material from the generating unit. For lines entering the load terminal, the branch implicit material flow density is equal to the consumption or emission of material from the generation side caused by the transmission of one unit of electricity. This has a clear physical meaning.

The branch implicit material flow is a non-directional scalar and a non-negative value, with dimensions identical to the material consumption or emission intensity at the generation side or generating unit. It quantifies the amount of material consumed or emitted on the generation side due to the transmission of one unit of electricity through the transmission branch.

2.2.4. Node implicit material intensity

The branch implicit material flow density describes the relationship between power flow and implicit material flow. Since the generation and consumption processes in the power system are represented as nodes, it is necessary to define a physical quantity that describes the intensity of implicit material at the node level. The node implicit material intensity, denoted by e , represents the implicit material intensity at node n , e_n , and is expressed as:

$$e_n = \frac{\sum_{i \in N} P_i \rho_i}{\sum_{i \in N} P_i} = \frac{\sum_{i \in N} v_i}{\sum_{i \in N} P_i} \quad (3)$$

In eq. (3), N represents the set of all branches connected to node n through which current flows into node n , and i denotes the branch number. The implicit material intensity at the node shares the same dimension as the implicit material flow density of the branches, numerically equating to the weighted average of the implicit material flow densities ρ_i of all branches flowing into node n with respect to the active power flow P_i . The physical significance of the node's implicit material intensity is the equivalent material consumption or emission on the generation side caused by consuming one unit of electricity at that node. Crucially, this intensity should not be viewed merely as a passive accounting output. Instead, it functions as a network-embedded responsibility "shadow signal". It reflects the virtual environmental responsibility borne by the node, serving as the foundational information carrier that bridges the physical source-side emissions with the economic load-side response. For a power plant node, its node implicit material intensity is equal to the real-time material emission or consumption intensity of the power plant's generation. This paper uses $e_{c,n}$ to denote the node's implicit carbon intensity, which shares the same dimension as the implicit carbon flow density, measured in $\text{ton}/(\text{MWh})$. Similarly, $e_{w,n}$ is used to denote the node's virtual water intensity, which shares the same dimension as the virtual water flow density, measured in $\text{m}^3/(\text{MWh})$.

In power systems, the consumption and emission of materials are primarily related to the active power output of generators, with very little influence from reactive power output. Therefore, the distribution of implicit material flows is mainly affected by the distribution of active power flows in the system.

Consider a system with N nodes, where K nodes are equipped with generator injections and M nodes are connected to loads, with the network topology (non-redundant node-branch incidence matrix) being known. The calculation of implicit material flows in the power system is derived from the results of the power flow analysis. The matrices used in this study for branch power distribution (P_B), generator injection distribution (P_G), load distribution (P_L), and active power flow (P_N) are defined as per reference [18].

Different generators have varying material consumption and emission intensities, which are known conditions in the calculation of implicit material flows, forming the generator material consumption and emission intensity vector of the system. Let the material consumption intensity of the k -th generator (where $k = 1, 2, \dots, K$) be e_{GK} , then the generator material consumption intensity vector can be represented as:

$$E_G = [e_{G1}, e_{G2}, \dots, e_{GK}]^T \quad (4)$$

The primary computation goal for implicit material flows in power systems is the implicit material intensity at all nodes. Let the implicit material intensity of the n -th node (where $n = 1, 2, \dots, N$) be e_n , then the vector of node implicit material intensities can be represented as:

$$E_N = [e_1, e_2, \dots, e_N]^T \quad (5)$$

The definition of the node implicit material intensity is given by eq. (3), which expresses the implicit material intensity e_i at node i in the system as:

$$e_i = \frac{\sum_{s \in S^+} P_{Bs} \rho_s + P_{Gi} e_{Gi}}{\sum_{s \in S^+} P_{Bs} + P_{Gi}} \quad (6)$$

The above equation can be rewritten in matrix form as:

$$e_i = \frac{\eta_N^{(i)} (P_B^T E_N + P_G^T E_G)}{\sum_{\nu=1}^N P_{B,\nu,i} + \sum_{t=1}^K P_{G,t,i}} \quad (7)$$

where $\eta_N^{(i)} = (0, 0, \dots, 1, \dots, 0)$ is an N -dimensional unit row vector, in which the i -th element is 1.

According to the definition of the node active power flux matrix, the equation can be obtained:

$$\sum_{\nu=1}^N P_{B,\nu,i} + \sum_{t=1}^K P_{G,t,i} = \eta_N^{(i)} P_N (\eta_N^{(i)})^T \quad (8)$$

Given eqs. (3) and (4) yields:

$$\eta_N^{(i)} P_N (\eta_N^{(i)})^T e_i = \eta_N^{(i)} (P_B^T E_N + P_G^T E_G) \quad (9)$$

Expanding eq. (9) to the full system dimension results:

$$P_N E_N = P_B^T E_N + P_G^T E_G \quad (10)$$

The calculation formula for the implicit material intensity at all nodes is:

$$E_N = (P_N - P_B^T)^{-1} P_G^T E_G \quad (11)$$

To ensure the invertibility of matrix $(P_N - P_B^T)$, the diagonal elements of the P_N matrix should be checked before calculation. If any zero elements are found, the corresponding node and the generators and lines connected to that node should be removed from the power grid. After updating the values of all calculation matrices and vectors, the system's implicit material flow distribution should be recalculated. The material flow calculation steps are shown in Fig. 2.

After calculating the node implicit material intensity vector, the implicit material flow rates R_B for each branch and the unit injection implicit material flow rates R_G can be further obtained as follows:

$$R_B = P_B \text{diag}(E_N) \quad (12)$$

$$R_G = P_G \text{diag}(e_{Gi}) \quad (13)$$

where e_{Gi} represents the implicit material intensity at the node corresponding to the location of the i -th unit.

2.3. Water resource stress

In order to reflect the differences in water resource endowment on the generator side, this paper adopts the water stress index (WSI) conversion method proposed by Pfister et al. [30]. This method uses a logistic function to transform the adjusted water availability index into a water stress index, with its normalized value ranging from 0.01 to 1.

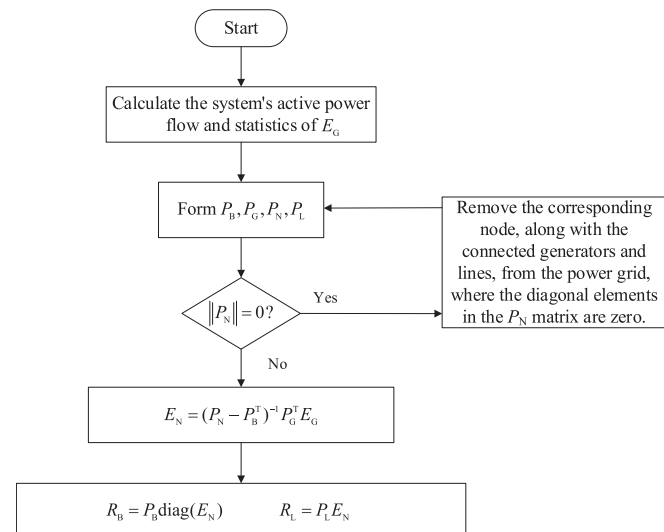


Fig. 2. Block diagram of power system material flow calculation.

Based on different water stress values, regions are classified into four categories of water scarcity:

1. Very Water-Rich Areas (WSI < 0.2)
2. Water-Rich Areas (0.2 ≤ WSI < 0.6)
3. Water-Scarce Areas (0.6 ≤ WSI < 0.8)
4. Severely Water-Scarce Areas (WSI > 0.8)

The calculation method for the water stress index (WSI) is as follows:

$$WTA = \frac{W_U - W_R - W_{\text{transfer}}}{R_W} \quad (14)$$

$$WSI = \frac{1}{1 + e^{-6.4 \cdot WTA \left(\frac{1}{0.01} - 1 \right)}} \quad (15)$$

where W_U is the total water use in a region, W_R is the available renewable water resources, and W_{transfer} is the net inflow of water.

3. Low-carbon and water-saving economic bi-level optimization scheduling scheme

3.1. Electricity-carbon-water Nexus responsibility sharing model based on node water and carbon intensities

At any given moment, the carbon intensity and water intensity at a load node determine the carbon emissions and water consumption per unit of electricity consumed. Based on this principle, optimal load-side scheduling can encourage electricity consumption when carbon and water intensities are low, and reduce consumption when these intensities are high. The nodal carbon and water intensities act as key mediators. By translating physical flow data into these shadow signals, the proposed model effectively links source-side generation scheduling with load-side demand response, ensuring that environmental responsibilities are allocated and managed efficiently across the nexus. This approach reduces carbon emissions and water consumption per unit of energy over a scheduling cycle, leading to effective energy conservation and emission reduction.

According to principles of consumer psychology, price-based demand response can guide users toward rational electricity consumption behavior by altering electricity prices. In China, significant progress has been made in guiding users to participate in peak shaving and valley filling through the establishment of time-of-use (TOU) pricing models. Referencing the TOU pricing model, this study analyzes from a low-carbon and water-saving perspective. Based on the node carbon and water intensities, low, medium, and high electricity-carbon-water coupling responsibility intervals are defined. A time-of-use electricity-carbon pricing model is established within each responsibility interval. The notations used throughout this chapter are summarized in Appendix.

Using the formulas to solve for the load node carbon intensity and water intensity, the current carbon and water intensities at each load node at time t are calculated. These are then used to classify the low, medium, and high electricity-carbon-water coupling responsibility intervals. The specific classification method of electricity-carbon-water coupling responsibility division is structured as follows:

$$e_{t,ni} \in \begin{cases} \text{Low responsibility, } e_{\min} \leq e_{t,i} \leq \frac{1}{2}(e_{\min} + e_{\text{ave}}) \\ \text{Medium responsibility, } \frac{1}{2}(e_{\min} + e_{\text{ave}}) < e_{t,i} \leq \frac{1}{2}(e_{\max} + e_{\text{ave}}) \\ \text{High responsibility, } \frac{1}{2}(e_{\max} + e_{\text{ave}}) < e_{t,i} \leq e_{\max} \end{cases} \quad (16)$$

$$w_{t,ni} \in \begin{cases} \text{Low responsibility, } w_{\min} \leq w_{t,i} \leq \frac{1}{2}(w_{\min} + w_{ave}) \\ \text{Medium responsibility, } \frac{1}{2}(w_{\min} + w_{ave}) < w_{t,i} \leq \frac{1}{2}(w_{\max} + w_{ave}) \\ \text{High responsibility, } \frac{1}{2}(w_{\max} + w_{ave}) < w_{t,i} \leq w_{\max} \end{cases} \quad (17)$$

In eq. (16), $e_{\min}$ and $e_{\max}$ represent the minimum and maximum node carbon intensities at load nodes at time t , e_{ave} represents the average node carbon intensity at load nodes at time t , $e_{t,i}$ represents the node carbon intensity at node i at time t . In eq. (17), $w_{\min}$ and $w_{\max}$ represent the minimum and maximum node water intensities at load nodes at time t , w_{ave} represent the average node water intensity at load nodes at time t , $w_{t,i}$ represent the node water intensity at node i at time t .

The low, medium, and high time-of-use carbon prices and virtual water prices with reference to peak, valley, and level time-of-use electricity prices are set by eqs. (18) and (19). On this basis, the calculation method of an electricity-carbon-water coupling price is obtained and presented in (20).

$$q_{co_2,t,i} = \begin{cases} q_0 - \frac{1}{2}\lambda, e_{t,i} \in \text{Low responsibility} \\ q_0, e_{t,i} \in \text{Medium responsibility} \\ q_0 + \frac{1}{2}\lambda, e_{t,i} \in \text{High responsibility} \end{cases} \quad (18)$$

$$q_{water,t,i} = \begin{cases} q_0 - \frac{1}{2}\mu, w_{t,i} \in \text{Low responsibility} \\ q_0, w_{t,i} \in \text{Medium responsibility} \\ q_0 + \frac{1}{2}\mu, w_{t,i} \in \text{High responsibility} \end{cases} \quad (19)$$

$$q_{cw,t} = \frac{q_{co_2,t,i} + q_{water,t,i}}{e_{t,i} + w_{t,i}} \quad (20)$$

In eqs. (18), λ represents the load carbon price difference for high, medium, and low responsibility intervals. This differential is derived from the CEA (Chinese Emission Allowance) price range of 50–120 ¥/tCO₂. By translating this to the power system context—multiplying by the average carbon emission intensity of the power system (approximately 0.6 tCO₂/MWh)—the range for λ is established as 30–72 ¥/MWh. In eqs. (19), μ represents the load water price difference for high, medium, and low responsibility intervals. As a market price or pricing basis for virtual water is currently unavailable, its value cannot be directly determined from market data. A synergy coefficient, α , is introduced to represent the relative importance of water conservation versus carbon reduction under current policy objectives. We set α within the interval 0.5–0.8, reflecting that the importance of the water conservation goal is defined as 50% to 80% of the carbon reduction goal. Applying this coefficient to the carbon price range 30–72 ¥/MWh yields a theoretical range for μ of 15–57.6 ¥/MWh. For ease of practical application, this is rounded to 15–60 ¥/MWh. q_0 represents the basic electricity price. The baseline electricity price q_0 is set at 500 ¥/MWh. $q_{co_2,t,i}$ represents the electricity-carbon coupling price for load i , $q_{water,t,i}$ represents the electricity-water coupling price for load i , and the electricity-carbon-water coupling price is represented by $q_{cw,t}$.

3.2. Bi-level scheduling model considering low-carbon and water-saving economic

The load aggregator is defined as an intermediary between the provider and buyer of demand response resources, which can be seen as the traditional distribution company or the load management center within a power grid company. The load.

aggregator clusters and analyzes user consumption characteristics and response capabilities, aggregating controllable resources to participate in the grid's scheduling efforts, thereby fulfilling the multiple objectives of the grid company, load aggregator, and end users, and maximizing the overall benefits.

After calculating the carbon intensity and water intensity at each node using the carbon emission flow theory and virtual water flow theory, each load aggregator (LA) receives the carbon and water intensity signals from the nodes. It then uses incentive-based electricity pricing contracts to guide flexible loads to respond. A flexible load carbon emission model based on node carbon intensity is constructed, which calculates the carbon emissions of flexible loads after demand response. Flexible loads can be categorized according to user response characteristics into transferable loads (TL) and curtailable loads (CL). This paper assumes that users will respond to price signals in a fully rational manner, and it uses this assumption as the foundation for implementing optimal dispatch strategies [31,32].

The source-load-storage coordinated low-carbon and water-saving economic bi-level scheduling framework is shown in Fig. 3. The upper and lower layers are coupled through the electricity-carbon-water price signals and load consumption demand signals. The upper-layer grid operator adjusts the power generation output plan by calculating generation costs, carbon trading costs, and virtual water trading costs. Based on the implicit material flow theory, the grid operator calculates the node carbon intensity and node water intensity, and transmits the node carbon intensity, water intensity, and time-of-use electricity prices to the lower-layer load aggregators (LA) and energy storage operators at different nodes.

Once the lower-layer LA receives the node carbon intensity and node water intensity signals, it takes on the corresponding carbon emission and water consumption responsibilities. Through incentive contracts, the LA adjusts flexible load consumption plans, and the electricity purchase demand is fed back to the upper-layer grid operator. Upon receiving the updated electricity purchase demand from the lower layer, the upper layer further adjusts the generation output and updates the node carbon intensity and node water intensity.

3.2.1. Upper layer grid operator economic scheduling

The upper layer aims to minimize the total cost of the grid operator, adjusting the power generation scheduling of the units. The total cost includes the generation costs of each power plant, carbon trading costs, and virtual water trading costs. The carbon trading cost is the product of the carbon emission intensity coefficient of the unit and its electricity generation. The virtual water trading cost is the product of the unit's water consumption intensity coefficient, regional water stress index, and its electricity generation. It converts the physical scarcity of water resources (WSI) into economic costs through the water rights trading mechanism. WSI, as a multiplier, captures the variation in the value of water rights under regional differences, while electricity generation represents the scale of economic activity, ensuring that costs are linked to trade volume. In regions with a high WSI, the transaction costs of electricity production are high, which encourages virtual water imports, reduces local water stress, and aligns with the sustainable goals of the water rights system [33]. The upper layer optimization model sets a series of constraints to ensure the stable and economical operation of the system.

Firstly, the output constraints of thermal power units and ramp rate constraints ensure the stable operation of thermal power plants. It effectively prevents the risks caused by output fluctuations and

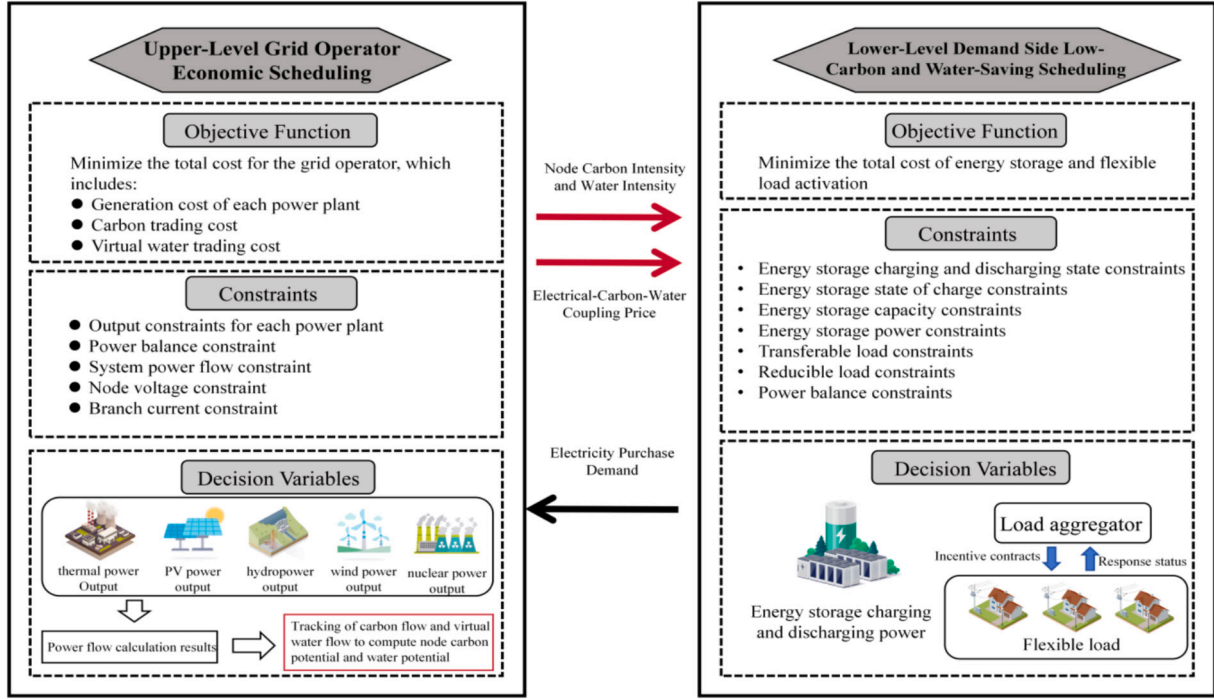


Fig. 3. Decision structure of the proposed bi-level optimal scheduling based on electricity-carbon-water nexus.

excessive ramping. Afterward, operational constraints for hydropower, nuclear power, wind power, and photovoltaic units are set to ensure the upper and lower limits of unit outputs. According to the power balance at the flow nodes, the electricity generated by the power plants equals the load consumption plus the transmission power through the lines. Accordingly, the system flow constraints are set to ensure the rationality and safety of power flow within the power grid.

3.2.1.1. Objective function. The objective function is the total operational cost of the grid operator, including the generation cost of each power plant F_G , carbon trading costs $F_{grid}^{CO_2}$, and virtual water trading costs F_{grid}^{water} . The goal of the upper layer economic scheduling is to minimize the total cost:

$$\min F = \min(F_G + F_{grid}^{CO_2} + F_{grid}^{water}) \quad (21)$$

$$F_G = \sum_{t=1}^T \sum_{g=1}^{N_w} C_e P_{g,t}^w + q_w (P_{g,t}^{wy} - P_{g,t}^w) \Delta t + \sum_{t=1}^T \sum_{p=1}^{N_{pv}} C_e P_{p,t}^{pv} + q_{pv} (P_{p,t}^{pvy} - P_{p,t}^{pv}) \Delta t \\ + \sum_{t=1}^T \sum_{i=1}^{N_G} C_e P_{i,t}^G + \sum_{t=1}^T \sum_{q=1}^{N_h} C_e P_{q,t}^h + \sum_{t=1}^T \sum_{k=1}^{N_n} C_e P_{k,t}^n \quad (22)$$

In eq. (22), N_w represents the number of wind power units; C_e represents the electricity purchase price from the power plants to the grid operator; $P_{g,t}^w$ represents the output of the g -th wind turbine at time t ; $P_{g,t}^{wy}$ represents the forecasted output of the g -th wind turbine at time t ; q_w represents the wind curtailment penalty coefficient for the g -th wind turbine; N_{pv} represents the number of photovoltaic (PV) units; $P_{p,t}^{pv}$ represents the output of the p -th photovoltaic unit at time t ; $P_{p,t}^{pvy}$ represents the forecasted output of the p -th photovoltaic unit at time t ; q_{pv} represents the photovoltaic curtailment penalty coefficient for the p -th photovoltaic unit; N_G represents the number of thermal power plants; $P_{i,t}^G$ represents the output of the i -th thermal power unit at time t ; N_h represents the number of hydroelectricity power plants; $P_{q,t}^h$ represents the output of the q -th hydroelectricity power unit at time t ; N_n represents the number of nuclear

power plants; $P_{k,t}^n$ represents the output of the k -th nuclear power unit at time t ;

$$F_{grid}^{CO_2} = C_{CO_2} \left(\sum_{t=1}^T \sum_{i=1}^{N_G} e^G P_{i,t}^G + \sum_{t=1}^T \sum_{q=1}^{N_h} e^h P_{q,t}^h + \sum_{t=1}^T \sum_{k=1}^{N_n} e^n P_{k,t}^n + \sum_{t=1}^T \sum_{g=1}^{N_w} e^w P_{g,t}^w + \sum_{t=1}^T \sum_{p=1}^{N_{pv}} e^{pv} P_{p,t}^{pv} \right) \Delta t \quad (23)$$

where e^G, e^h, e^n, e^w , and e^{pv} represent the carbon emission coefficient of thermal, hydropower, nuclear power, wind power and photovoltaic power, respectively; C_{CO_2} represents carbon emission cost.

$$F_{grid}^{water} = C_w \left(\sum_{t=1}^T \sum_{i=1}^{N_G} WSI_n w^G P_{i,t}^G + \sum_{t=1}^T \sum_{q=1}^{N_h} WSI_n w^h P_{q,t}^h + \sum_{t=1}^T \sum_{k=1}^{N_n} WSI_n w^n P_{k,t}^n + \sum_{t=1}^T \sum_{g=1}^{N_w} WSI_n w^w P_{g,t}^w + \sum_{t=1}^T \sum_{p=1}^{N_{pv}} WSI_n w^{pv} P_{p,t}^{pv} \right) \Delta t \quad (24)$$

where w^G, w^h, w^n, w^w , and w^{pv} represent the water consumption coefficient of thermal, hydropower, nuclear power, wind power and photovoltaic power, respectively; C_w represents water consumption cost. The virtual water trading cost C_w accounts for the economic implications of water consumption in electricity generation, adjusted for regional water scarcity. WSI_n represents the water pressure index of node n . Consider a thermal power unit at a node with $WSI_n = 0.9$ (severely water-scarce region) generating $P_{i,t}^G = 100 \text{ MWh}$ at time t , with water consumption coefficient $w^G = 3.32 \text{ m}^3/\text{MWh}$ and $C_w = 1.2 \text{ ¥/m}^3$. The physical virtual water consumed is $3.32 \times 100 = 332 \text{ m}^3$, adjusted by WSI to reflect scarcity: $332 \times 0.9 = 298.8 \text{ m}^3$ equivalent scarcity-adjusted water. The trading cost is then $298.8 \times 1.2 = 358.56 \text{ ¥}$. In contrast, for a wind unit at the same node, the cost would be only $0.56 \times 100 \times 0.9 \times 1.2 \approx 60.48 \text{ ¥}$. The formula economically penalizes high-water generation in scarce areas, encouraging a shift to low-water renewables and aligning with policy goals for sustainable resource allocation. In regions with a high WSI , the transaction costs of electricity production are high, which

encourages virtual water imports, reduces local water stress, and aligns with the sustainable goals of the water rights system.

3.2.1.2. Constraint conditions. To account for potential forecasting errors in renewable generation, prediction error coefficients k^w and k^{pv} are introduced in the wind and photovoltaic constraints (25) and (26). These coefficients dynamically adjust upper bounds to form uncertainty intervals.

(1) Wind Power Constraints

$$\begin{cases} 0 \leq P_{g,t}^w \leq (1 + k^w)P_{g,t}^{wy} \\ 0 \leq |k^w| \leq k_{\max}^w \end{cases} \quad (25)$$

(2) Photovoltaic power constraints

$$\begin{cases} 0 \leq P_{p,t}^{pv} \leq (1 + k^{pv})P_{p,t}^{py} \\ 0 \leq |k^{pv}| \leq k_{\max}^{pv} \end{cases} \quad (26)$$

(3) Thermal Power Generation Constraints

a. Output Constraint

$$P_{i,\min}^G \leq P_{i,t}^G \leq P_{i,\max}^G \quad (27)$$

$$Q_{i,\min}^G \leq Q_{i,t}^G \leq Q_{i,\max}^G \quad (28)$$

where $P_{i,\max}^G$ and $P_{i,\min}^G$ represent the maximum and minimum active power output of the i -th thermal power unit, respectively; $Q_{i,\max}^G$ and $Q_{i,\min}^G$ represent the maximum and minimum reactive power of the i -th thermal power unit, respectively.

b. Ramp-up constraint

$$\begin{cases} P_{i,t}^G - P_{i,t-1}^G \leq R_i^U P_i^{G,\max} \\ P_{i,t-1}^G - P_{i,t}^G \leq R_i^D P_i^{G,\max} \end{cases} \quad (29)$$

where $P_{i,t-1}^G$ is the output of the i -th thermal power unit at time $t-1$; R_i^U and R_i^D represent the maximum ramp-up rate and maximum ramp-down rate of the i -th thermal power unit, respectively.

(4) Hydropower Generation Constraints

a. Output Constraint

$$P_{q,\min}^h \leq P_{q,t}^h \leq P_{q,\max}^h \quad (30)$$

$$Q_{q,\min}^h \leq Q_{q,t}^h \leq Q_{q,\max}^h \quad (31)$$

where $P_{q,\max}^h$ and $P_{q,\min}^h$ represent the maximum and minimum active power output of the q -th hydropower unit, respectively; $Q_{q,\max}^h$ and $Q_{q,\min}^h$ represent the maximum and minimum reactive power of the q -th hydropower unit, respectively.

b. Ramp-up constraint

$$\begin{cases} P_{q,t}^h - P_{q,t-1}^h \leq R_q^U P_q^{h,\max} \\ P_{q,t-1}^h - P_{q,t}^h \leq R_q^D P_q^{h,\max} \end{cases} \quad (32)$$

where $P_{q,t-1}^h$ is the output of the q -th hydropower unit at time $t-1$; R_q^U and R_q^D represent the maximum ramp-up rate and maximum ramp-down rate of the q -th thermal power unit, respectively.

(5) Nuclear Power Constraints.

a. Output Constraint

$$P_{k,\min}^n \leq P_{k,t}^n \leq P_{k,\max}^n \quad (33)$$

$$Q_{k,\min}^n \leq Q_{k,t}^n \leq Q_{k,\max}^n \quad (34)$$

where $P_{k,\max}^n$ and $P_{k,\min}^n$ represent the maximum and minimum active

power output of the k -th nuclear power unit, respectively; $Q_{k,\max}^n$ and $Q_{k,\min}^n$ represent the maximum and minimum reactive power output of the k -th nuclear power unit, respectively.

b. Ramp-up constraint

$$\begin{cases} P_{k,t}^n - P_{k,t-1}^n \leq R_k^U P_k^{n,\max} \\ P_{k,t-1}^n - P_{k,t}^n \leq R_k^D P_k^{n,\max} \end{cases} \quad (35)$$

$P_{k,t-1}^n$ is the output of the k -th nuclear power unit at time $t-1$; R_k^U and R_k^D represent the maximum ramp-up rate and maximum ramp-down rate of the k -th nuclear power unit, respectively.

(6) Power Flow Constraints

a. Power Balance Constraint

$$p_{j,t}^{\text{load}} = \text{load}_{j,t} - p_{\text{CL},t} + p_{\text{TL},t} + \sum_{i=1}^{N_{\text{ess}}} P_{\text{cha},i,t} - \sum_{i=1}^{N_{\text{ess}}} P_{\text{dis},i,t} \quad (36)$$

where $p_{j,t}^{\text{load}}$ represents the active power load at node j at time t ; $\text{load}_{j,t}$ represents the forecasted load at node j at time t ; $p_{\text{CL},t}$ represents the transferable load at node j at time t ; $p_{\text{TL},t}$ represents the curtailable load at node j at time t ; $P_{\text{cha},i,t}$ represents the charging power of the i -th energy storage system at time t ; $P_{\text{dis},i,t}$ represents the discharging power of the i -th energy storage system at time t ; N_{ess} represents the number of energy storage systems.

$$\sum_{i \in V_{G,j}} P_{i,t}^G + \sum_{g \in V_{W,j}} P_{g,t}^W + \sum_{p \in V_{PV,j}} P_{p,t}^{pv} + \sum_{q \in V_{H,j}} P_{q,t}^h + \sum_{k \in V_{N,j}} P_{k,t}^n - \sum_{m \in V_{e,j}} P_{m,t} = p_{j,t}^{\text{load}} \quad (37)$$

where $V_{G,j}$, $V_{W,j}$, $V_{PV,j}$, $V_{H,j}$ and $V_{N,j}$ represent the set of thermal plants, wind power plants, photovoltaic plants power plants, hydropower plants and nuclear power plants at node j , respectively; $V_{e,j}$ represents the set of transmission lines connected to the node j ; $P_{m,t}$ represents the active power flowing into branch j .

$$\sum_{i \in V_{G,j}} Q_{i,t}^G + \sum_{q \in V_{H,j}} Q_{q,t}^h + \sum_{k \in V_{N,j}} Q_{k,t}^n - \sum_{m \in V_{e,j}} Q_{m,t} = Q_{j,t}^{\text{load}} \quad (38)$$

where $Q_{m,t}$ represents the reactive power flowing into branch j ; $Q_{j,t}^{\text{load}}$ represents the reactive power load at node j at time t .

b. System Power Flow Constraints

$$V_{j,t} = V_{i,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2)I_{ij,t} \quad (39)$$

$$I_{ij,t} = \frac{P_{ij,t}^2 + Q_{ij,t}^2}{V_{i,t}} \quad (40)$$

where $V_{j,t}$ is the square of the voltage at the terminal node j of the branch at time t ; $V_{i,t}$ is the square of the voltage at the starting node i of the branch at time t ; r_{ij} and x_{ij} are the resistance and reactance of the line; $P_{ij,t}$ and $Q_{ij,t}$ are the active and reactive power transmitted on the line at time t , and $I_{ij,t}$ is the current transmitted on the line. To accurately characterize the physical characteristics of the distribution network, this paper adopts the DistFlow model. Since the aforementioned DistFlow equations are non-convex, directly embedding them into the optimization model would result in a challenging non-convex programming problem that is difficult to solve. To efficiently solve the model while preserving accuracy, we employ second-order cone relaxation (SOCR) techniques, which transform the non-convex power flow constraints into an exact or relaxed equivalent in the form of a SOCP model.

c. Node voltage constraint

$$V_j^{\min} \leq V_{j,t} \leq V_j^{\max} \quad (41)$$

$$V_i^{\min} \leq V_{i,t} \leq V_i^{\max} \quad (42)$$

where $V_j^{\max}$ and $V_j^{\min}$ represent the upper and lower limits of the voltage magnitude at node j ; $V_i^{\max}$ and $V_i^{\min}$ represent the upper and lower limits of the voltage magnitude at node i .

d. Branch current constraint.

$$0 \leq I_{ij,t} \leq I_{ij}^{\max} \quad (43)$$

where $I_{ij}^{\max}$ is the maximum allowable current that can be transmitted through the line.

3.2.2. Lower-level demand-side low-carbon water-saving scheduling

The lower-level aims to minimize the total cost of energy storage and load scheduling in response to the carbon and water intensity signals from the upper-level. It uses the electricity-carbon-water coupled pricing model to guide users and energy storage systems in participating in low-carbon and water-saving responses, adjusting flexible load consumption and energy storage charging and discharging strategies.

3.2.2.1. *Objective function.* 1) The lower-level demand-side low-carbon and water-saving scheduling cost f includes the following components:

2) The purchase cost of electricity from the upper-level by the Load Aggregator (LA) C^{buy} ;

3) The demand response subsidy cost for Curtailable Load (CL) and Transferable Load (TL) C^{ct} ;

4) The energy storage system scheduling cost C^{ess} .

$$\min f = C^{\text{buy}} + C^{\text{ct}} + C^{\text{ess}} \quad (44)$$

$$C^{\text{buy}} = \sum_{t=1}^T q_{\text{cw},t} P_{\text{load},t} \Delta t \quad (45)$$

$$C^{\text{ct}} = C_{\text{de}} \sum_{t=1}^T (p_{\text{CL},t} + |U_{1,t}| p_{\text{TL},t}) \Delta t \quad (46)$$

where $U_{1,t}$ represents the demand response state of user 1 at moment t .

$$C^{\text{ess}} = e_{\text{ess}} \sum_{t=1}^T \left(\sum_{i=1}^{N_{\text{ess}}} P_{\text{cha},i,t} + \sum_{i=1}^{N_{\text{ess}}} P_{\text{dis},i,t} \right) \Delta t \quad (47)$$

where e_{ess} represents the energy storage scheduling cost per unit.

3.2.2.2. Constraint conditions.

(1) Energy Storage Constraints

a. Energy Storage State Constraints

$$X_{\text{cha},i,t} + X_{\text{dis},i,t} \leq 1 \quad (48)$$

where $X_{\text{cha},i,t}$ and $X_{\text{dis},i,t}$ represent the charging and discharging status of the energy storage device, respectively, with binary values of 0 and 1.

b. State of Charge (SOC) Constraint

$$SOC_i^{\min} \leq SOC_{i,t} \leq SOC_i^{\max} : \epsilon_{i,1,t}^{\text{SOC,L}}, \epsilon_{i,1,t}^{\text{SOC,U}} \quad (49)$$

$$SOC_{i,t} = \frac{E_{\text{ess},i,t}}{E_{\text{bat},i}} \quad (50)$$

where $SOC_i^{\max}$ and $SOC_i^{\min}$ represent the upper and lower limits of the state of charge (SOC) of the i -th energy storage system, respectively; $SOC_{i,t}$ is the state of charge of the i -th energy storage system at time t ; $E_{\text{ess},i,t}$ is the energy of the i -th energy storage system at time t ; $E_{\text{bat},i}$ is the rated capacity of the i -th energy storage system. Note that λ_i is the dual variable of the equality constraint, and ϵ_i is the dual variable of the inequality constraint.

c. Energy Storage Power Constraints

$$\begin{cases} 0 \leq P_{\text{cha},i,t} \leq X_{\text{cha},i,t} P_{\text{cha},i}^{\max} : \epsilon_{i,2,t}^{\text{cha,L}}, \epsilon_{i,2,t}^{\text{cha,U}} \\ 0 \leq P_{\text{dis},i,t} \leq X_{\text{dis},i,t} P_{\text{dis},i}^{\max} : \epsilon_{i,3,t}^{\text{dis,L}}, \epsilon_{i,3,t}^{\text{dis,U}} \end{cases} \quad (51)$$

where $P_{\text{cha},i}^{\max}$ and $P_{\text{dis},i}^{\max}$ are the upper limits of the charging and discharging power for the i -th energy storage system.

d. Energy Storage Capacity Constraint

$$E_{\text{ess},i,t} = E_{\text{ess},i,t-1} + \eta X_{\text{cha},i,t} P_{\text{cha},i,t} - \eta X_{\text{dis},i,t} P_{\text{dis},i,t} : \lambda_{i,1,t}^{\text{ess}} \quad (52)$$

where η is the charge/discharge efficiency; $E_{\text{ess},i,t-1}$ is the energy level of the i -th energy storage system at time $t-1$.

(2) Flexible load constraints.

a. Transferable load constraint

$$\sum_{t=1}^T |U_{1,t}| \leq M_1 : \epsilon_{i,4,t}^{\text{U,L}} \quad (53)$$

where M_1 represents the maximum number of response times for user 1 within a cycle.

b. Curtailable Load Constraints

$$0 \leq p_{\text{CL},t} \leq P_{\text{CL},t}^{\max} : \epsilon_{i,5,t}^{\text{CL,L}}, \epsilon_{i,5,t}^{\text{CL,U}} \quad (54)$$

where $P_{\text{CL},t}^{\max}$ is the maximum curtailable load.

c. Load Balance Constraint

$$P_{\text{load},t} - p_{\text{CL},t} + p_{\text{TL},t} + \sum_{i=1}^{N_{\text{ess}}} P_{\text{cha},i,t} - \sum_{i=1}^{N_{\text{ess}}} P_{\text{dis},i,t} = P_{\text{load},t} : \lambda_{i,2,t}^{\text{Elec}} \quad (55)$$

where $p_{\text{load},t}$ represents the load condition of the upper layer at time t .

3.3. Bi-level optimization solution

The low-carbon, water-saving economic dual-level optimization scheduling model considering source-load-storage coordination proposed in this paper is a mathematical optimization problem with equilibrium constraints [34]. The bi-level model is coupled and cannot be solved directly. Therefore, the lower-level optimization problem is replaced by its KKT optimality conditions, converting the original bi-level model into a single-level mathematical program with equality constraints, inequality constraints, and complementarity slackness conditions. The presence of complementarity slackness conditions renders the transformed model a non-convex problem. To handle the nonlinear complementarity constraints without relying on conventional approaches such as the big-M method or SOS1 techniques, a branch-and-bound algorithm is implemented. This method systematically enumerates and prunes the solution space, ensuring global optimality while avoiding issues associated with big-M bounds [35].

The branch-and-bound algorithm operates through the following key steps and is shown in Fig. 4:

a. Relaxation: At each node, a relaxed version of the problem is solved by temporarily ignoring the complementarity conditions, yielding a tractable convex problem that provides a lower bound on the objective.

b. Bounding: A global upper bound is maintained from the best feasible solution found, while node-specific lower bounds are obtained from the relaxed solutions.

c. Branching: If the relaxed solution violates complementarity constraints, the space is partitioned by branching on specific complementarity pairs.

d. Pruning: Nodes are pruned if their lower bound exceeds the current global upper bound, eliminating suboptimal subspaces.

This iterative process narrows the optimality gap until convergence, guaranteeing a global optimum with advantages in robustness and completeness for non-convex complementarity problems. The MATLAB

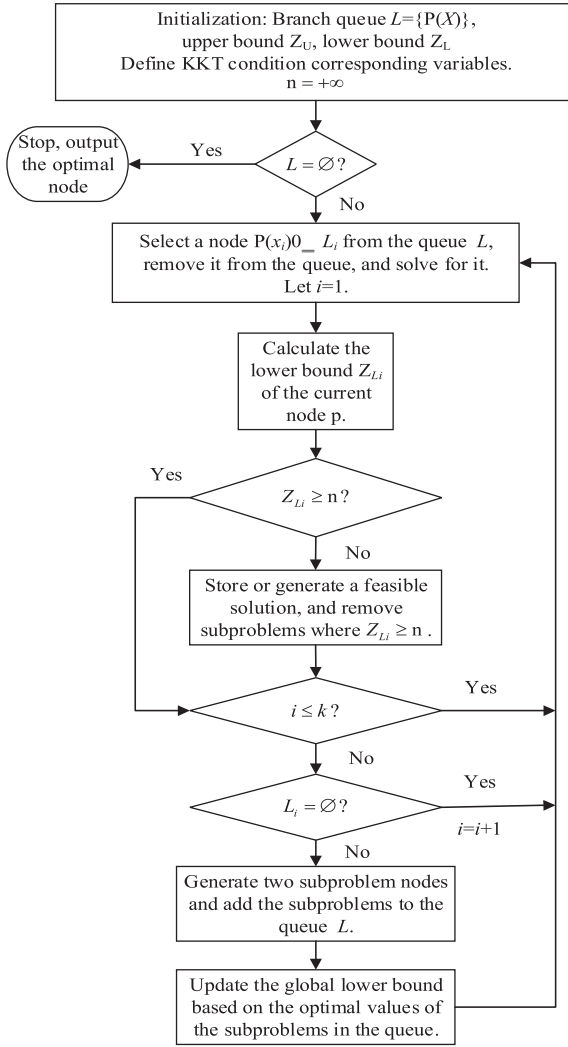


Fig. 4. Branch-and-bound solution process.

CPLEX solver is employed to solve the relaxed subproblems within the branch-and-bound framework.

The lower-level demand-side low-carbon water-saving scheduling model is equivalent to the KKT conditions, as shown in eqs. (56) to (68). Eq. (56) represents the Lagrange function obtained from the lower-level objective function, eqs. (57) to (63) represent the first-order partial derivatives of the lower-level model's Lagrange function, and eqs. (64) to (68) represent the complementary slackness conditions in the KKT conditions.

$$L = f + \sum_{e \in \Omega_{\text{inequal}}} \varepsilon_{i,e,t} h_{i,e,t} + \sum_{k \in \Omega_{\text{equal}}} \lambda_{i,k,t} g_{i,k,t} \quad (56)$$

$$\frac{\partial L}{\partial P_{\text{load},t}} = \sum_{t=1}^T q_{\text{cw},t} \Delta t - \lambda_{i,2,t}^{\text{Elec}} = 0 \quad (57)$$

$$\frac{\partial L}{\partial P_{\text{CL},t}} = \sum_{t=1}^T C_{\text{de}} \Delta t - \lambda_{i,2,t}^{\text{Elec}} + \varepsilon_{i,5,t}^{\text{CL,L}} - \varepsilon_{i,5,t}^{\text{CL,U}} = 0 \quad (58)$$

$$\frac{\partial L}{\partial P_{\text{TL},t}} = \sum_{t=1}^T C_{\text{de}} |U_{1,t}| \Delta t + \lambda_{i,2,t}^{\text{Elec}} = 0 \quad (59)$$

$$\frac{\partial L}{\partial U_{1,t}} = \sum_{t=1}^T C_{\text{de}} P_{\text{TL},t} \left(\frac{\partial |U_{1,t}|}{\partial U_{1,t}} \right) \Delta t - \varepsilon_{i,4,t}^{\text{U,L}} + \sum_{t=1}^T \left(\frac{\partial |U_{1,t}|}{\partial U_{1,t}} \right) = 0 \quad (60)$$

$$\frac{\partial L}{\partial P_{\text{cha},i,t}} = \sum_{t=1}^T \sum_{i=1}^{N_{\text{ess}}} e_{\text{ess}} \Delta t - \lambda_{i,1,t}^{\text{ess}} \eta + \sum_{i=1}^{N_{\text{ess}}} \lambda_{i,2,t}^{\text{Elec}} + \varepsilon_{i,2,t}^{\text{cha,L}} - \varepsilon_{i,2,t}^{\text{cha,U}} = 0 \quad (61)$$

$$\frac{\partial L}{\partial P_{\text{dis},i,t}} = \sum_{t=1}^T \sum_{i=1}^{N_{\text{ess}}} e_{\text{ess}} \Delta t + \lambda_{i,1,t}^{\text{ess}} \eta - \sum_{i=1}^{N_{\text{ess}}} \lambda_{i,2,t}^{\text{Elec}} + \varepsilon_{i,3,t}^{\text{dis,L}} - \varepsilon_{i,3,t}^{\text{dis,U}} = 0 \quad (62)$$

$$\frac{\partial L}{\partial \text{SOC}_{i,t}} = \varepsilon_{i,1,t}^{\text{SOC,L}} - \varepsilon_{i,1,t}^{\text{SOC,U}} = 0 \quad (63)$$

$$\begin{cases} 0 \leq \varepsilon_{i,1,t}^{\text{SOC,L}} \perp \text{SOC}_{i,t} - \text{SOC}_i^{\min} \geq 0 \\ 0 \leq \varepsilon_{i,1,t}^{\text{SOC,U}} \perp \text{SOC}_i^{\max} - \text{SOC}_{i,t} \geq 0 \end{cases} \quad (64)$$

$$\begin{cases} 0 \leq \varepsilon_{i,2,t}^{\text{cha,L}} \perp P_{\text{cha},i,t} \geq 0 \\ 0 \leq \varepsilon_{i,2,t}^{\text{cha,U}} \perp X_{\text{cha},i,t} P_{\text{cha},i,t}^{\max} - P_{\text{cha},i,t} \geq 0 \end{cases} \quad (65)$$

$$\begin{cases} 0 \leq \varepsilon_{i,3,t}^{\text{dis,L}} \perp P_{\text{dis},i,t} \geq 0 \\ 0 \leq \varepsilon_{i,3,t}^{\text{dis,U}} \perp X_{\text{dis},i,t} P_{\text{dis},i,t}^{\max} - P_{\text{dis},i,t} \geq 0 \end{cases} \quad (66)$$

$$0 \leq \varepsilon_{i,4,t}^{\text{U,L}} \perp M_t - \sum_{t=1}^T |U_{1,t}| \geq 0 \quad (67)$$

$$\begin{cases} 0 \leq \varepsilon_{i,5,t}^{\text{CL,L}} \perp p_{\text{CL},i,t} \geq 0 \\ 0 \leq \varepsilon_{i,5,t}^{\text{CL,U}} \perp p_{\text{CL},i,t}^{\max} - p_{\text{CL},i,t} \geq 0 \end{cases} \quad (68)$$

4. Case study analysis

4.1. Test system description

This study uses an improved IEEE-33 node model, as presented in Fig. 5, for 24-h simulations. The IEEE 33 system is a standard test feeder for distribution network analysis, operating at a base voltage of 12.66 kV and a system capacity of 100 MVA [40]. The line parameters of the IEEE 33 system are shown in the Appendix (for lines r and x). Node 1 is connected to thermal power with a maximum output of 2 MW; Node 2 is connected to hydroelectric power with a maximum output of 0.8 MW; Node 3 is connected to nuclear power with a maximum output of 1 MW; Node 16 is connected to distributed photovoltaic (PV), and Node 18 is connected to distributed wind power. The predicted outputs of PV and wind power are shown in Fig. 11. and Fig. 12. For this study, the assumption $k^w = k^{\text{PV}} = 0$ maintains deterministic optimization, consistent with baseline forecasting. Future extensions may set non-zero values for stochastic analysis. The carbon emission intensity and water consumption intensity of the unit refer to Reference [36]. The energy storage system is connected at Node 17, and the storage parameters are shown in Table 1. The load aggregator can choose three shiftable loads and two reducible loads for control. The shiftable loads are located at Nodes 4, 15, and 30, and the reducible loads are located at Nodes 10 and 26. The maximum dispatchable load is 30% of the load at the control nodes.

The improved IEEE-33 node model assumed that node 1 is located in an extremely water-scarce region, node 2 is located in an extremely water-rich region, node 3 is in a water-abundant region, node 16 is in an extremely water-scarce region, and node 18 is in a water-scarce region. Assuming the carbon emission cost is 252 ¥/tCO₂, the water consumption cost is 5 ¥/m³, and the penalty coefficients for wind and solar curtailment is 120 ¥/MWh. To verify the effectiveness of the proposed bi-level optimal scheduling method based on electricity-carbon-water nexus, particularly in comparison with recently published methods, such as traditional economic scheduling and low-carbon optimization scheduling, the test cases of the three simulation scenarios are conducted, as shown in Table 2.

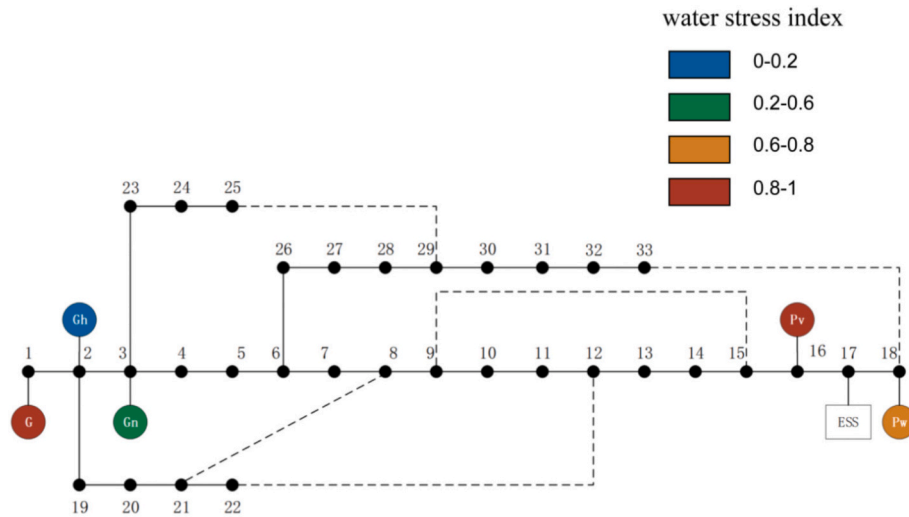


Fig. 5. Improved IEEE-33 node model.

Table 1
Energy Storage Parameters.

Energy Storage	Rated Capacity/MWh	Maximum Charging and Discharging Power/MW	Charging and Discharging Efficiency	State of Charge Range
ESS1	1.03	0.375	90%	10%–90%
ESS2	1.21	0.375	90%	10%–90%

Table 2
Test cases in three scenarios.

Cases	Economic	Low-carbon	Water-saving
Scenario 1	✓	×	×
Scenario 2	✓	✓	×
Scenario 3	✓	✓	✓

In Table 1, only economic scheduling is considered in Scenario 1. It simulates the traditional economic scheduling model, primarily focusing on operational costs, serving as a baseline for comparison with other scenarios. In Scenario 2, the economic and low-carbon scheduling are considered while ignoring the feature of the water saving. Scenario 2 is configured to serve as a comparative baseline against Scenario 3. The goal is to assess whether the proposed bi-level optimal scheduling

method, based on the electricity-carbon-water nexus, yields better results in low-carbon and water-saving optimization than conventional low-carbon optimization models [37–39]. Scenario 3 not only covers economic efficiency but also incorporates low-carbon and water-saving objectives. As a result, it offers a more comprehensive scheduling scheme, which is not addressed by current published methods. Through the analysis of Scenario 3, the model's unique advantages in coordinating water resources and carbon emissions are revealed.

4.2. Analysis of carbon emissions from power systems

The 24-h load node carbon intensity for three scenarios is shown in Fig. 6. The surface change trends for all three scenarios are generally consistent, with two peak periods. The first peak in node carbon intensity occurs around hour 11, marking the highest peak of the entire period. After hour 13, the overall node carbon intensity starts to decline, reaching a peak-to-valley point around hour 16 before gradually rising again. It ultimately reaches a minor peak around hour 20 before declining. The overall trend of node carbon intensity follows a pattern consistent with the system's load variations.

From Table 2, it can be seen that the total system 24-h carbon emissions in Scenario 2 and Scenario 3 have decreased by 8.43% and 10.68%, respectively, compared to Scenario 1, indicating significant carbon reduction effects in both scenarios. The total carbon emissions in Scenario 3 are 4.85% lower than those in Scenario 2, suggesting that LA

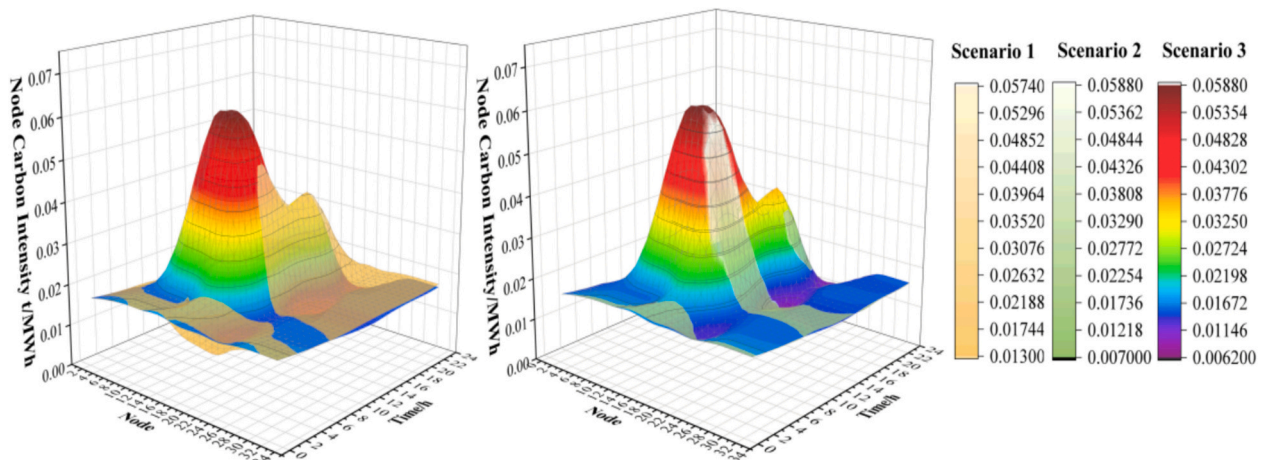


Fig. 6. 24-h Load Node Carbon intensity for Scenarios 1–3.

(Load Aggregator) participation in low-carbon scheduling can further enhance the carbon reduction effect to some extent.

From the node carbon intensity distribution map, it can be observed that the best carbon reduction effect occurs in Scenario 3, where the carbon reduction mainly takes place during the periods when the node carbon intensity is not at its highest peak, compared to Scenario 1. In contrast, in Scenario 3 compared to Scenario 2, the main carbon reduction occurs during the periods when the node carbon intensity is at its highest peak.

Node 1 consistently has the highest average node carbon intensity among all three scenarios. Fig. 7 depicts the node carbon intensity of node 1 for three different scenarios. The results showcase that, in Scenario 1, Node 1's carbon intensity remains above 1 t/MWh after the 5th period. However, in Scenario 2 and Scenario 3, the average node carbon intensity of Node 1 decreases significantly, with carbon intensity noticeably reduced in periods 1–3 and 9–17 compared to Scenario 1. The carbon intensity variation trends in Node 1 are nearly identical between Scenario 2 and Scenario 3, with only a small difference in the average node carbon intensity. This suggests that coal-fired power plants, which are both high carbon and high-water consumption units, have a synergistic effect on water and carbon reduction at specific nodes.

Node 1 hosts the high carbon-emitting coal-fired power unit G. As demonstrated in Table 3, the total carbon flow rate from the coal-fired power units in Scenarios 2 and 3 is notably lower than in Scenario 1. This outcome highlights the effectiveness of the low-carbon optimization scheduling process in curbing the output of high-carbon-emitting units. These results underscore the proposed method's advantage in achieving substantial carbon reductions, showcasing its superiority over conventional economic scheduling approaches that do not prioritize environmental objectives.

4.3. Analysis of virtual water consumed from power systems

The 24-h load node water intensity for three different scenarios is shown in Fig. 8. Observably, there are significant differences in the overall trends of node water intensity between Scenario 1 and Scenarios 2 and 3. In Scenario 1, the node water intensity remains at a relatively high level starting from period 3. In contrast, Scenarios 2 and 3 exhibit a bimodal trend similar to the node carbon intensity. The first peak of node water intensity in Scenarios 2 and 3 appears around period 7. After dropping to the trough around period 13, it begins to rebound and rise, reaching the second peak around period 19 before gradually decreasing. Compared to the trend of node carbon intensity, the node water intensity peaks 2–3 h earlier, indicating that water-saving scheduling and carbon reduction scheduling have temporal differences. In Scenarios 2 and 3, compared to Scenario 1, the node carbon intensity and node water

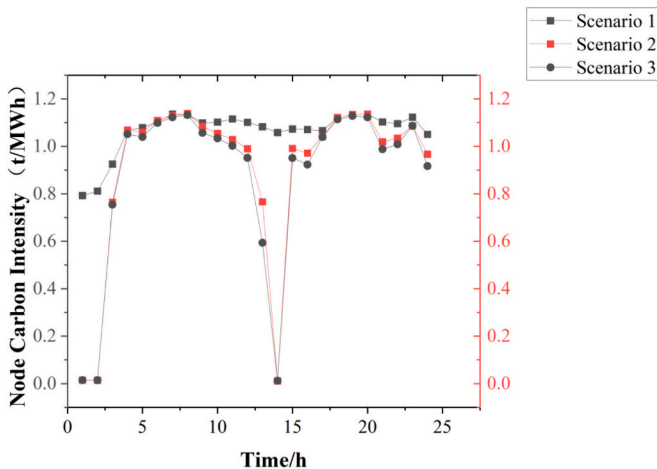


Fig. 7. The node 1 carbon intensity for scenarios 1–3.

intensity for nodes 19–22 both show a downward trend, with a more significant decrease in node water intensity, suggesting that water-saving scheduling and carbon reduction scheduling also have spatial differences.

In order to study the variation of virtual in the nexus, Table 4 depicts the total water flow rate injected by each unit. It can be seen that the total water consumption of the system in Scenarios 2 and 3 decreased by 23.62% and 29.64%, respectively, compared to Scenario 1. Although Scenario 2 does not explicitly consider water-saving demand, it still achieves a notable water-saving effect. The significant 23.62% water-saving effect in Scenario 2 is attributed to a strong positive correlation between carbon emission intensity and water consumption intensity. High-carbon thermal units, such as the G unit at Node 1, typically exhibit high water consumption, whereas low-carbon wind and photovoltaic units, like the PV at Node 16 and WP at Node 18, have minimal water use. Consequently, when Scenario 2 prioritizes carbon reduction by dispatching renewable energy and curtailing thermal power, water consumption is reduced as a byproduct, resulting in an inherent synergy without explicit water pricing signals.

To explicitly characterize the underlying synergy between water conservation and carbon reduction, rather than reporting them in parallel, a quantitative synergy indicator is introduced: *WCR* (Water Saving per Unit Carbon Reduction). This metric is calculated as the ratio of reduced water consumption to reduced carbon emissions relative to the baseline (Scenario 1). It is calculated as:

$$WCR = \frac{\Delta W}{\Delta C} = \frac{W_{S1} - W_{Sk}}{C_{S1} - C_{Sk}} \quad (k = 2, 3) \quad (69)$$

where *W* represents the total water consumption (m^3) and *C* represents the total carbon emissions (t). Subscripts S1, S2, and S3 denote Scenario 1, 2, and 3, respectively.

Based on the data in Tables 3 and 4, we calculate the *WCR* for Scenarios 2 and 3 as follows:

a. For Scenario 2:

The water consumption decreased from $578.86m^3$ to $442.15m^3$ ($\Delta W = 136.71m^3$), and carbon emissions decreased from 3.119 t to 2.865 t ($\Delta C = 0.254t$).

$$WCR = \frac{136.71}{0.254} \approx 538.2m^3 / tCO_2 \quad (70)$$

b. For Scenario 3:

The water consumption decreased to $407.24m^3$ ($\Delta W = 171.62m^3$), and carbon emissions decreased to 2.786 t ($\Delta C = 0.333t$).

$$WCR = \frac{171.62}{0.333} \approx 515.5m^3 / tCO_2 \quad (71)$$

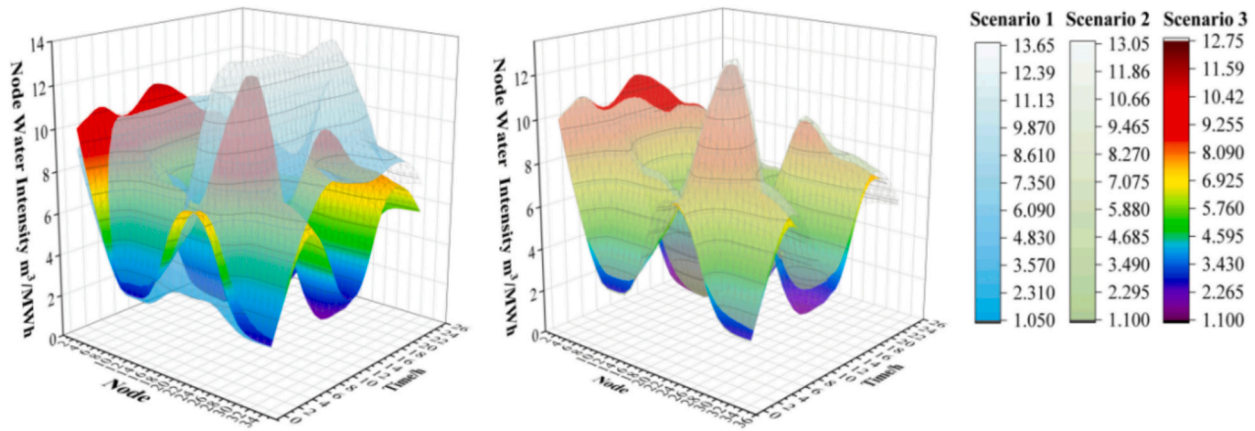
The higher *WCR* value in Scenario 2 reveals a strong intrinsic synergy between water and carbon. This is because high-carbon thermal units are typically also high-water consumers. Consequently, prioritizing carbon reduction naturally leads to significant water savings as a byproduct.

However, Scenario 3 achieves a lower ratio but yields the highest absolute reductions in both water and carbon. This indicates that while the intrinsic physical correlation is beneficial, the proposed coupling model provides superior absolute performance by navigating trade-offs in specific contexts. For instance, in water-scarce regions like Node 1, Scenario 3 uses water pricing signals to suppress high-water generation more aggressively than Scenario 2, which focuses solely on carbon intensity. Therefore, this indicator verifies that the proposed integrated optimization model maximizes the comprehensive environmental benefits, rather than merely relying on the inherent correlation between

Table 3

Total Carbon Flow Rate Injected by Each Generator and Total System Carbon Emissions for Scenarios 1–3.

Scenarios	G Total Carbon Flow Rate Injected (tCO ₂ /h)	Gh Total Carbon Flow Rate Injected (tCO ₂ /h)	Gn Total Carbon Flow Rate Injected (tCO ₂ /h)	Pw Total Carbon Flow Rate Injected (tCO ₂ /h)	Pv Total Carbon Flow Rate Injected (tCO ₂ /h)	Total System Carbon Consumption (t)
1	21.693	0.248	0.410	0.220	0.320	3.119
2	15.989	0.253	0.410	0.305	0.515	2.865
3	14.237	0.253	0.410	0.298	0.515	2.786

**Fig. 8.** The 24-h node water intensity for scenarios 1–3.**Table 4**

Total water flow rate injected by each unit and total water consumption of the system for scenarios 1–3.

Scenario	G Injected Total Water Flow Rate (m ³ /h)	Gh Injected Total Water Flow Rate (m ³ /h)	Gn Injected Total Water Flow Rate (m ³ /h)	Pw Injected Total Water Flow Rate (m ³ /h)	Pv Injected Total Water Flow Rate (m ³ /h)	Total
1	58.553	334.609	74.400	2.660	7.091	578.86
2	43.157	341.760	74.400	3.677	11.403	442.15
3	38.427	341.250	74.400	3.606	11.403	407.24

resources.

In Scenario 3, compared to Scenario 2, the overall trend of node water intensity is consistent, further indicating that water-saving and carbon reduction have a certain level of synergy. The main advantage of Scenario 3 is its capability for precise regulation of generation in water-stressed regions. Node 1 is situated in an extremely water-scarce area with a Water Stress Index of 0.9. In Scenario 2, which lacks water price signals, the high-water-consumption thermal unit at Node 1 might still be dispatched. In contrast, Scenario 3 introduces a water price differential, which substantially increases the comprehensive generation cost at Node 1, encompassing generation, carbon, and high-water costs. This cost escalation strongly incentivizes the model to suppress the output of the G unit, as evidenced by its much lower water flow injection rate in Scenario 3 compared to Scenario 2. The system addresses this by importing more virtual water from water-rich regions like Node 2 or by utilizing local low-water renewable resources, thereby directly alleviating local water stress. The total water consumption in Scenario 3 decreased by 7.9% compared to Scenario 2, demonstrating that considering water-saving optimization in Scenario 3 results in superior water-saving performance compared to Scenario 2.

Furthermore, the inclusion of the water price differential enables Scenario 3 to navigate temporal trade-offs between different power sources. For example, it can differentiate between a low-carbon but high-water source like run-of-river hydropower and a zero-carbon, zero-

water source like photovoltaics. During periods of high load or water scarcity, Scenario 3 prioritizes PV based on nodal water intensity, simultaneously achieving superior water savings and fulfilling carbon reduction objectives. This mechanism optimizes the generation mix not only for carbon efficiency but also for water conservation in a context-sensitive manner.

When analyzing the water-saving effects in different scenarios, it is important to consider the water scarcity level at different locations, which distinguishes it from carbon emission analysis. Node 1, where the thermal power plant G is located, is an extremely water-scarce area. Scenario 3 has the lowest total water flow rate among the three scenarios, indicating that Scenario 3 plays a significant role in alleviating water pressure in the region around Node 1. Node 16, where the photovoltaic power station is located, is also in an extremely water-scarce area. The main carbon emissions and water consumption for photovoltaics during their entire lifecycle come from the production and transportation of photovoltaic equipment, making their carbon emissions and water consumption intensity relatively low, and their impact on local water resources is minimal. Therefore, increasing the absorption of photovoltaic power during the scheduling process plays a key role in achieving water-saving and carbon-reduction for the overall system. Node 2, which has the highest water intensity, shows that, as seen in Fig. 9, the node water intensity in Scenario 1 remains consistently high without significant changes. The node water intensity in Scenarios 2 and 3 follows a similar trend to the overall system node water intensity, with the variation in total water flow rate from hydropower being minor. This is because Node 2 is located in an area with abundant water, so the demand for water-saving scheduling in this region is relatively low.

The results indicate that the proposed bi-level optimization scheduling model for low-carbon and water-saving economic scheduling, which integrates source-load-storage coordination, effectively reduces system carbon emissions and alleviates regional water resource shortages. It achieves a dual reduction in water consumption and carbon emissions per unit of electricity consumed. In Scenario 3, the incorporation of demand-side response scheduling based on node carbon and water intensity leads to further reductions in total system carbon emissions and water consumption compared to Scenario 2. These

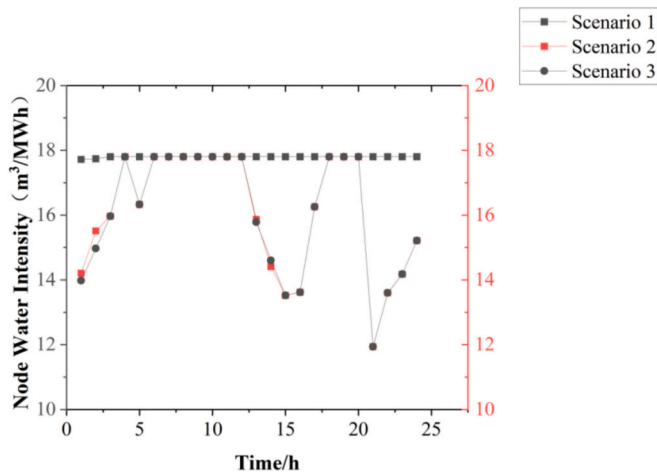


Fig. 9. The node 2 water intensity for scenarios 1–3.

findings highlight the superior performance of the proposed optimization model in achieving comprehensive environmental benefits, demonstrating its distinct advantage over existing methods in balancing carbon reduction and water conservation.

4.4. Analysis of economic performance

The comparison of power system scheduling costs for the three scenarios is shown in Table 5. In the three scenarios, Scenario 3 has the lowest total cost, with Scenario 2 following closely. Scenario 2 shows a 5.16% decrease in total cost compared to Scenario 1, primarily due to reductions in carbon emission costs and gains from the demand-side electricity-carbon coupling subsidy. By leveraging demand-side response and energy storage, Scenario 2 shifts high-carbon potential loads to times with lower carbon intensity, enhancing the system's intake of clean energy.

Compared to Scenarios 1 and 2, Scenario 3 achieves total cost reductions of 8.82% and 3.87%, respectively. This is because Scenario 3 takes a comprehensive approach to water and carbon reduction, considering the operational traits of the source-load-storage system. It prioritizes power generation from units with superior water-saving and low-carbon attributes and maximizes the flexibility of responsive loads and energy storage to transfer or cut high-carbon and high-water potential loads. By boosting the responsiveness of flexible loads, Scenario 3 lessens reliance on the power source, leading to more significant cuts in carbon emissions and water usage.

4.5. Analysis of electricity-carbon-water coupling responsibility division

As shown in Fig. 10, the 24-h distribution of the electricity-carbon-water coupling price at each node is presented. The electricity-carbon-water coupling price is formulated based on the carbon and water intensities, dividing the water-saving and carbon-reduction responsibility zones. A positive coupling price indicates the cost that the lower-level system needs to bear, while a negative value represents a subsidy cost, guiding the lower-level demand-side response and energy storage

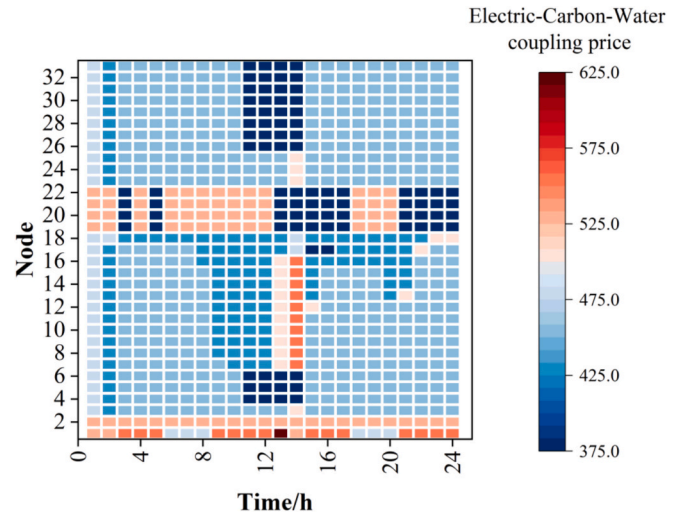


Fig. 10. Distribution of electricity-carbon-water coupling prices.

charging/discharging.

From the distribution of the electricity-carbon-water coupling prices at each node, the different carbon reduction and water-saving responsibilities at various load nodes can be observed. In the Fig. 10, nodes 1 to 2 are the main high-responsibility nodes for water-saving and carbon-reduction, where they are mostly in the high-responsibility zones for water-saving and carbon-reduction. Nodes 19 to 22 are in high-responsibility periods for water-saving and carbon-reduction during the time intervals of 6–12 and 18–20, which generally align with the peak periods of the system's average carbon and water intensities. Nodes 7 to 16 show high responsibility for water-saving and carbon-reduction during the 13–14 time period, which corresponds to the peak of the system's carbon and water intensities.

The highest water-saving and carbon-reduction responsibility across the entire system occurs at node 1 during the 13th time period, which aligns closely with the peak of the system's carbon and water intensities, further proving the accuracy of the electricity-carbon-water coupling price model.

4.6. Analysis of clean energy consumption in the power system

Fig. 11 and Fig. 12 depict the system's consumption of PV and wind power. As shown in Fig. 11 and Fig. 12, scenario 1 mainly focuses on minimizing the system's economic cost, which leads to higher curtailment of both photovoltaic and wind power. This is because the economic cost of coal-fired power plants must be considered, and no demand response is introduced in Scenario 1. As a result, the high-carbon and high-water-consuming coal-fired plants have higher output, resulting in higher carbon emissions and water consumption for the system.

In contrast, Scenario 2 takes into account the low-carbon benefits of the source-load-storage coordination, significantly improving the system's ability to absorb photovoltaic and wind power while reducing dependence on high-carbon coal-fired power plants. Scenario 3, based on Scenario 2, incorporates water-saving benefits from source-load-storage coordination. The photovoltaic and wind power absorption

Table 5
System operating costs for three scenarios.

Scenario	Generator Power Generation Cost	Demand Side Activation Cost	Energy Storage Activation Cost	Carbon Trading Cost	Virtual Water Trading Cost	Total Cost
1	49,207	×	1400	7046	802	58,455
2	49,395	−135	−816	6169	830	55,443
3	48,169	−331	−1033	5691	803	53,299

Note that × presents the unconsidered cost for different scenarios.

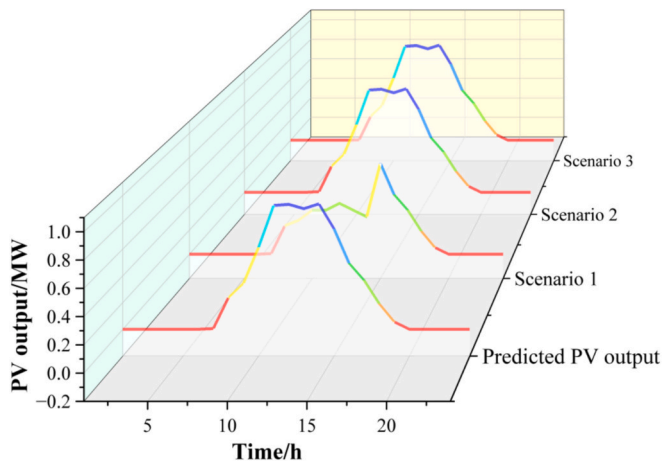


Fig. 11. Comparison of photovoltaic consumption for the three scenarios.

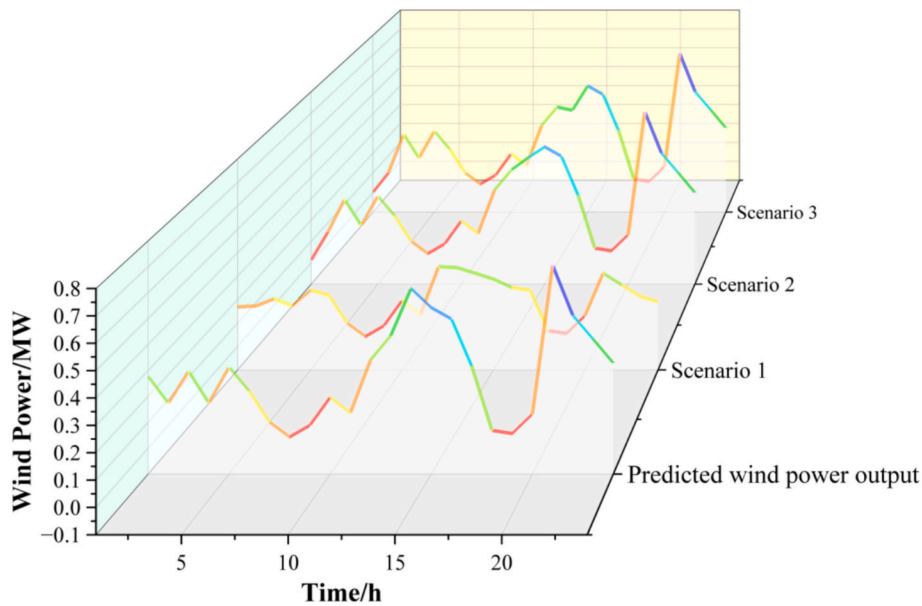


Fig. 12. Comparison of wind power consumption for the three scenarios.

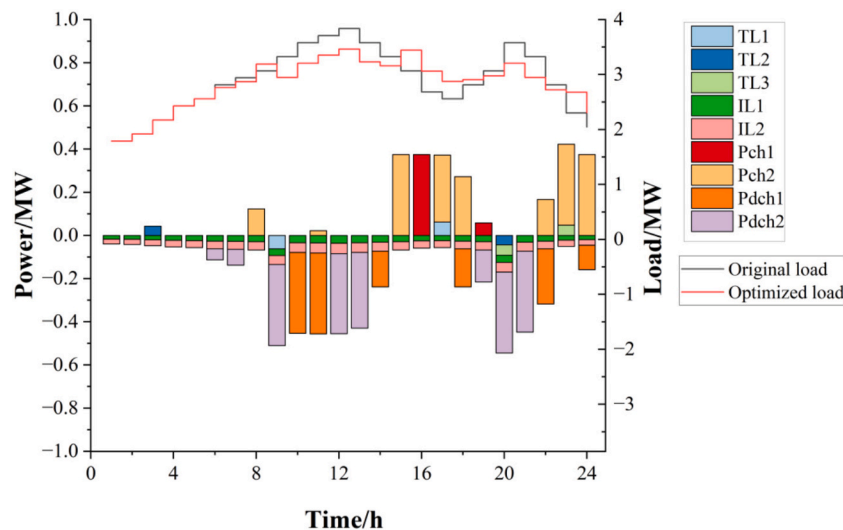


Fig. 13. Flexible load and energy storage activation conditions in Scenario 3.

remains similar to Scenario 2, but Scenario 3 further reduces the system's carbon emissions and redistributes the water flow, alleviating regional water scarcity issues.

4.7. Analysis of energy storage and flexible load scheduling

The flexible load and energy storage activation in Scenario 3 and the comparison of load curves before and after optimization are presented in Fig. 13. The original load curve has a large peak-to-valley difference. After implementing demand-side response and energy storage scheduling, the load curve exhibits a “peak shaving and valley filling” effect. From the Fig. 13, it can be seen that the distribution of the system's node carbon intensity and water intensity is consistent with the trend of the load curve.

Therefore, after the demand-side response and energy storage scheduling, the overall node carbon intensity and water intensity also experience a “peak shaving and valley filling” effect.

Specifically, transferable load TL1 is transferred to increase during the low-carbon and low-water intensity periods at node 3, and reduced

during the high-carbon and high-water intensity period at node 20. The same applies to transferable loads TL2 and TL3. Curtailable loads IL1 and IL2 are reduced in each time period, but the reduction is larger during peak periods and smaller during valley periods. The dual energy storage systems are responsible for charging and discharging tasks. When the state of charge of one energy storage unit reaches its constraint, the storage device switches between charging and discharging modes.

The energy storage system's neighboring nodes are the photovoltaic units at node 16 and the wind power units at node 18. The carbon intensity and water intensity at the energy storage node are influenced by these neighboring nodes, which in turn affects the energy storage charging and discharging. From the wind power output curve, it can be seen that the energy storage charging and discharging trends are similar to the wind power output curve. When the wind power and photovoltaic output increase, the carbon intensity and water intensity at the energy storage node decrease, and the energy storage enters charging mode to absorb the low-carbon and water-saving electricity from wind and photovoltaic power. When wind and photovoltaic outputs decrease, the energy storage enters discharging mode to release the stored clean energy, which helps reduce the output from high-carbon and high-water-consuming units, achieving carbon reduction and water-saving effects.

In summary, the proposed bi-level optimization scheduling model demonstrates clear advantages in effectively balancing carbon reduction and water conservation, outperforming existing methods that often neglect the integrated impact of carbon and water resource management [37,38]. By comprehensively considering source-load-storage coordination and incorporating demand-side responses, this model addresses the limitations of traditional approaches, which typically focus solely on economic scheduling or carbon reduction without adequately accounting for water resource constraints.

5. Conclusion

Research on water conservation and carbon reduction in power systems is of significant practical importance for alleviating regional water resource pressures and mitigating the global greenhouse effect. Based on the theory of carbon emission flow in power systems, this paper expands the theoretical research objects and applies the theory to the study of other material consumption and emissions in power systems, establishing the concept of the implicit material flow theory for power systems. By accurately solving carbon emission flow and virtual water flow data in power systems based on the network topology, system power flow distribution, and unit-side carbon emission and water consumption information, it enables the tracking and analysis of hidden carbon and virtual water flows in power networks from both temporal and spatial dimensions, providing a theoretical foundation for the electricity-carbon-water coupled optimization scheduling of power systems.

A coupled model of electricity, carbon, and water based on load node carbon intensity and water intensity is proposed, and a time-varying electricity-carbon-water coupled price model is established. Based on this, a coordination optimization scheduling strategy considering the source-load-storage (SLS) coordination is designed, and a bi-level low-carbon, water-saving, economic scheduling model is developed. The simulation results show that the responsibilities for carbon emissions and water consumption are transmitted from the unit side to the load side through the electricity-carbon-water coupled model in the form of node carbon intensity and node water intensity. The changes in node carbon intensity and node water intensity can be analyzed from both time and space dimensions, allowing for the accurate calculation of carbon emissions and water consumption from load node electricity consumption.

The method demonstrates highest effectiveness in power systems characterized by significant spatial heterogeneity of water resources, where the transfer of virtual water can alleviate regional scarcity.

Furthermore, the model relies on the availability of flexible resources; a higher proportion of energy storage and responsive loads enhances the system's ability to respond to the coupling price signals, thereby amplifying the water-carbon synergy. Conversely, in regions with uniform water abundance or systems with severe rigid constraints and minimal flexibility, the marginal benefits of the complex coupling mechanism may be less pronounced.

The study indicates that the proposed model, when compared to scenarios focusing solely on economic dispatch, can lead to a 10.68% decrease in carbon emissions and a 29.64% reduction in water usage. Moreover, when contrasted with scenarios that only account for low-carbon economic dispatch, the model can further reduce carbon emissions by 4.85% and water consumption by 7.9%. This model notably enhances the integration of new energy sources. Specifically, it increases photovoltaic energy consumption by 38.8% and wind power consumption by 25.7%, compared to scenarios that only consider economic dispatch. This reduction in reliance on thermal power units with high carbon emissions and water usage achieves a synergistic objective of low-carbon water conservation and energy structure optimization for the system.

The proposed bi-level optimization scheduling model for low-carbon, water-saving, and economic operation, considering source-load-storage coordination, can achieve power balance on the source-load-storage sides of the power system, ensure safe and stable system operation, and simultaneously address the system's economy, low-carbon performance, and water conservation. While the model's assumption of rational responses to coupled price signals may appear idealized, it does not compromise the validity of our core conclusions, since the focus of the study lies in exploring system-level coordination and the interactions across electricity, carbon, and water flows. Incorporating more realistic behavioral elements, such as correction functions to account for user inertia or refined calibration of transferable loads (TL) and curtailable loads (CL), may enhance the robustness of the framework. We acknowledge that this is an important direction for future research, where techniques from behavioral economics and uncertainty modeling could be integrated to better reflect the diversity and bounded rationality of user responses in real-world scenarios.

The nodal water-carbon coupling price proposed in this study serves as a potential prototype for distributed environmental responsibility pricing. Unlike traditional electricity pricing which only reflects energy supply costs, or standard carbon markets which focus on generation-side caps, this coupling price internalizes the scarcity of water resources and carbon intensity directly into the consumption signal at specific network nodes. It acts as a complementary instrument to existing market mechanisms, guiding flexible resources to respond to localized environmental pressures. By embedding these externalities into the price signal, the framework supports a more granular and efficient energy-environment-economy system, paving the way for market designs that explicitly value the electricity-carbon-water nexus.

CRediT authorship contribution statement

Yiyi Zhang: Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Yilang Zhao:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Xianfu Lin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Nomenclature

Notations	Definitions
$e_{\min}, e_{\max}$	Minimum and maximum node carbon intensities at load nodes at time t
e_{ave}	Average node carbon intensity at load nodes at time t
$w_{\min}, w_{\max}$	Minimum and maximum node water intensities at load nodes at time t
w_{ave}	Node water intensity at node i at time t
λ	Load carbon price difference for high, medium, and low responsibility intervals
μ	Load water price difference for high, medium, and low responsibility intervals
q_0	Basic electricity price
$q_{co_2,t,i}$	Electricity-carbon coupling price for load i
$q_{water,t,i}$	Electricity-water coupling price for load i
$q_{cw,t,i}$	Electricity-carbon-water coupling price
N_w	Number of wind power units
C_e	Electricity purchase price from the power plants to the grid operator
$p_{g,t}^w$	Output of the g -th wind turbine at time t
$\hat{p}_{g,t}^w$	Forecasted output of the g -th wind turbine at time t
q_w	Wind curtailment penalty coefficient for the g -th wind turbine
N_{pv}	Number of photovoltaic (PV) units
$p_{p,t}^{pv}$	Output of the p -th photovoltaic unit at time t
$\hat{p}_{p,t}^{pv}$	Forecasted output of the p -th photovoltaic unit at time t
q_{pv}	Photovoltaic curtailment penalty coefficient for the p -th photovoltaic unit
N_G	Number of thermal power plants
$p_{i,t}^G$	Output of the i -th thermal power unit at time t
N_h	Number of hydroelectricity power plants
$p_{q,t}^h$	Output of the q -th hydroelectricity power unit at time t
N_n	Number of nuclear power plants
$p_{k,t}^n$	Output of the q -th hydroelectricity power unit at time t
e^G	Carbon emission coefficient of thermal power generation
e^h	Carbon emission coefficient of hydroelectricity power generation
e^n	Carbon emission coefficient of nuclear power generation
e^w	Carbon emission coefficient of wind power generation
e^{pv}	Carbon emission coefficient of photovoltaic power generation
w^G	Water consumption coefficient of thermal power generation
w^h	Water consumption coefficient of hydroelectricity power generation
w^n	Water consumption coefficient of nuclear power generation
w^w	Water consumption coefficient of wind power generation
w^{pv}	Water consumption coefficient of photovoltaic power generation
$V_{G,j}$	Set of thermal power plants at node j
$V_{w,j}$	Set of wind power plants at node j
$V_{pv,j}$	Set of photovoltaic power plants at node j
$V_{h,j}$	Set of hydropower plants at node j
$V_{n,j}$	Set of nuclear power plants at node j
$V_{e,j}$	Set of transmission lines connected to the node j
$P_{in,t}$	Active power flowing into branch j
$p_{j,t}^{pload}$	Active power load at node j at time t
$\hat{load}_{j,t}$	Forecasted load at node j at time t
$p_{CL,t}$	Curtailable load at node j at time t
$p_{TL,t}$	Transferable load at node j at time t
$P_{cha,j,t}$	Charging power of the i -th energy storage system at time t
$P_{dis,i,t}$	Discharging power of the i -th energy storage system at time t
N_{ess}	Number of energy storage systems
$Q_{in,t}$	Reactive power flowing into branch j
$Q_{j,t}^{pload}$	Reactive power load at node j at time t
$V_{j,t}$	Square of the voltage at the terminal node j of the branch at time t
$V_{i,t}$	Square of the voltage at the starting node i of the branch at time t
r_{ij}, x_{ij}	Resistance and reactance of the line
$P_{ij,t}, Q_{ij,t}$	Active and reactive power transmitted on the line at time t
$I_{ij,t}$	Current transmitted on the line
$P_{load,t}$	LA purchase of electricity from upper Layer
C_{de}	Customer load scheduling incentive pricing
$U_{i,t}$	Demand response state of user i at moment t
e_{ess}	Energy storage scheduling cost per unit
$SOC_{i,t}$	State of charge of the i -th energy storage system at time t
$E_{ess,i,t}$	Energy of the i -th energy storage system at time t
$E_{bat,i}$	Rated capacity of the i -th energy storage system

(continued on next page)

(continued)

Notations	Definitions
η	Charge/discharge efficiency
$P_{load,t}$	Load condition of the upper layer at time t

Appendix B. For lines r and x of the studied IEEE 33 system

Starting node	Terminal node	r (Ω)	x (Ω)
1	2	0.0922	0.0470
2	3	0.4930	0.2511
3	4	0.3660	0.1864
4	5	0.3811	0.1941
5	6	0.8190	0.7070
6	7	0.1872	0.6188
7	8	0.7114	0.2351
8	9	1.0300	0.7400
9	10	1.0440	0.7400
10	11	0.1966	0.0650
11	12	0.3744	0.1238
12	13	1.4680	1.1550
13	14	0.5416	0.7129
14	15	0.5910	0.5260
15	16	0.7463	0.5450
16	17	1.2890	1.7210
17	18	0.7320	0.5740
2	19	0.1640	0.1565
19	20	1.5042	1.3554
20	21	0.4095	0.4784
21	22	0.7089	0.9373
3	23	0.4512	0.3083
23	24	0.8980	0.7091
24	25	0.8960	0.7011
6	26	0.2030	0.1034
26	27	0.2842	0.1447
27	28	1.0590	0.9337
28	29	0.8042	0.7006
29	30	0.5075	0.2585
30	31	0.9744	0.9630
31	32	0.3105	0.3619
32	33	0.3410	0.5302

Data availability

No data was used for the research described in the article.

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